Lecture 21 (11/4)

ML vs. stats, Feature maps

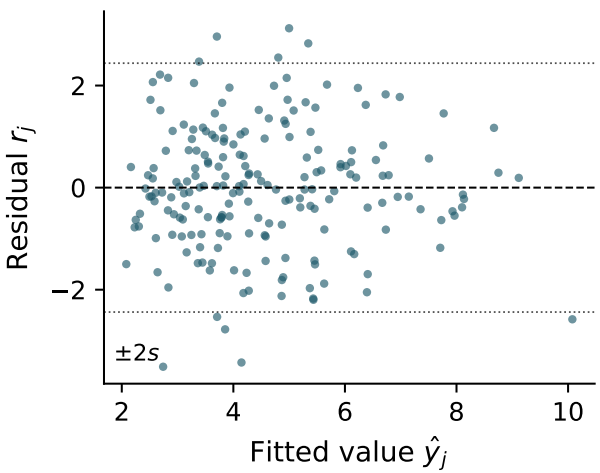
Today:

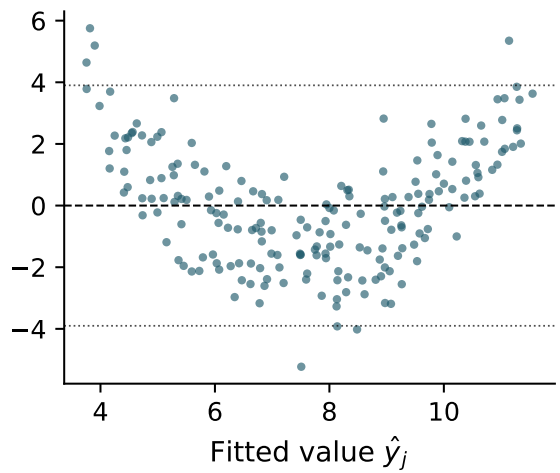
1) Residual plots: the spike protein
2) Statistics and machine learning
3) Feature maps and polynomial regression and overfitting
4) Midterm 2 return

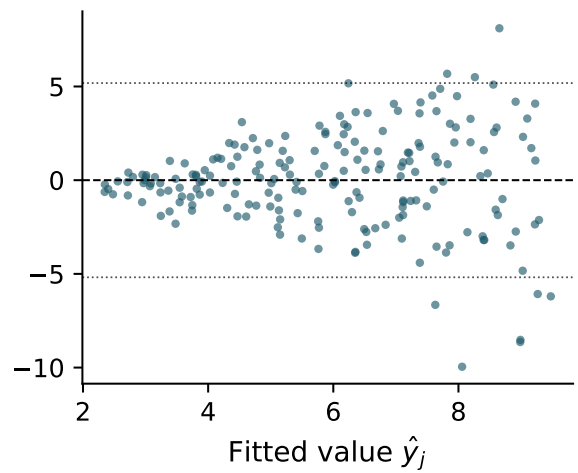
Recap: residuals

$r_j = y_j - \hat{y}_j$. So $r_j > 0$ means the model underestimates y_j, and $r_j < 0$ means it overestimates it.

The model is accurate when the residuals are symmetric about 0 in the vertical direction, few points fall outside the 95% interval, and the spread is the same at every $\hat{y}_j$.

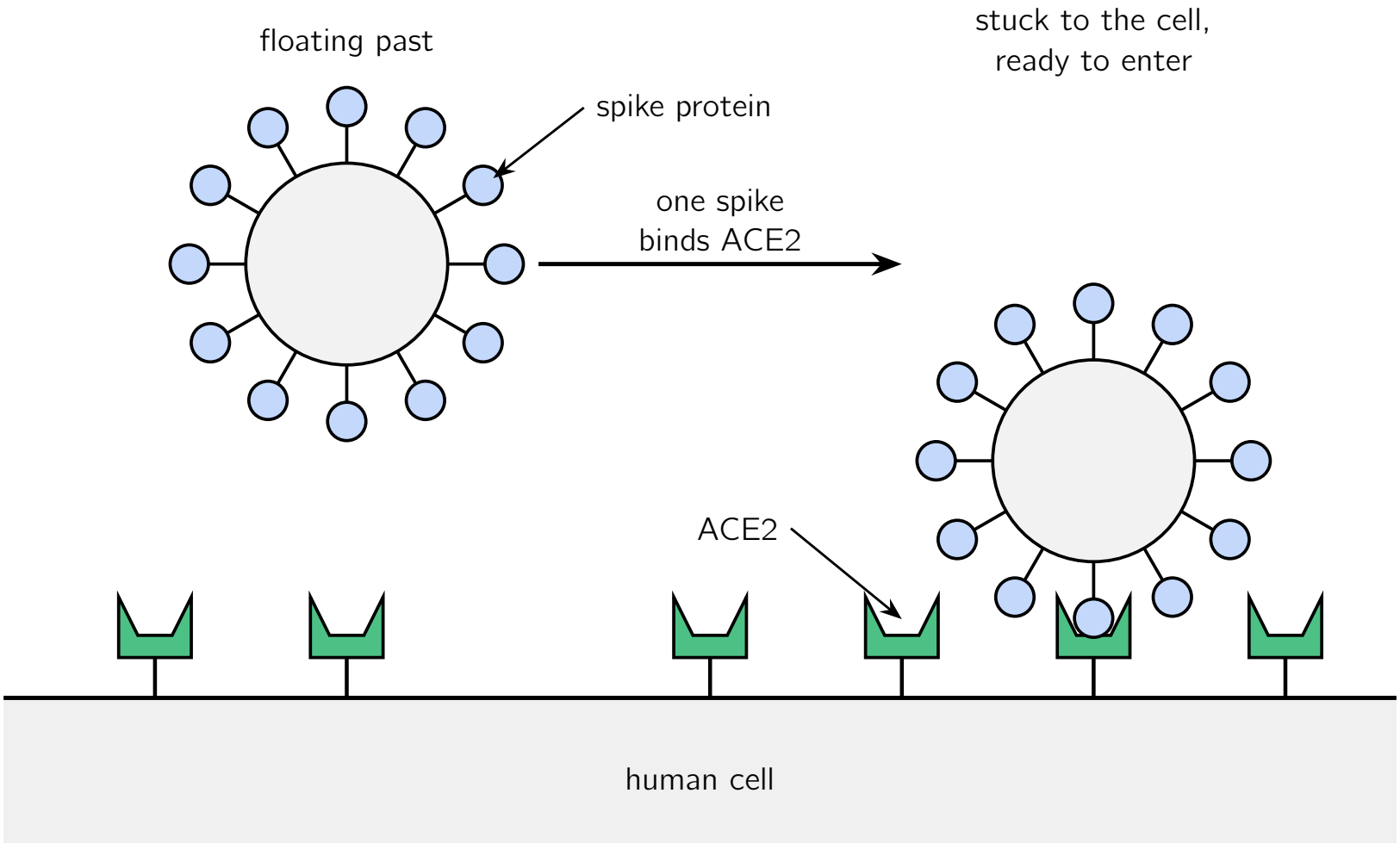






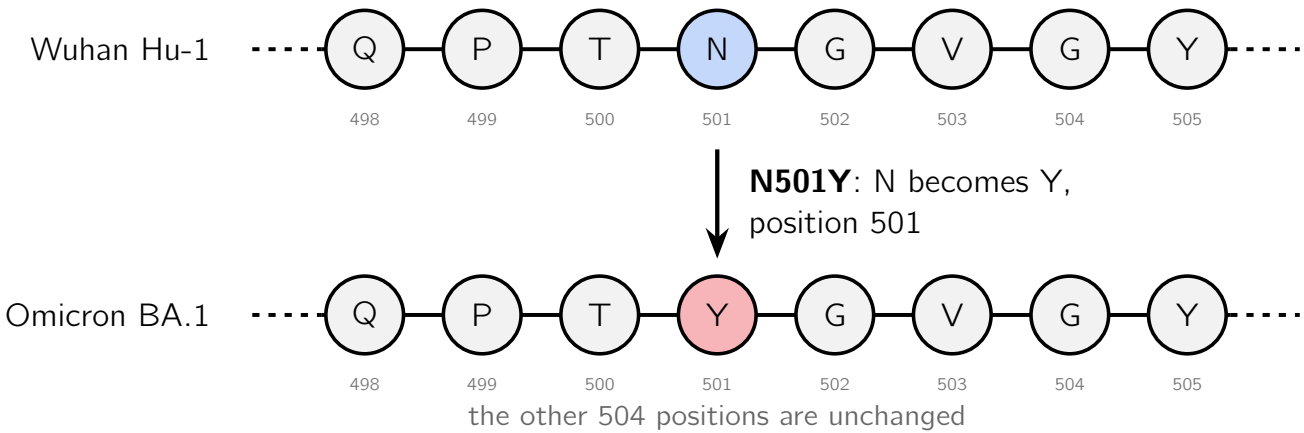
Background: the spike protein and ACE2

A virus gets into a cell by sticking to a protein on its surface. For SARS-CoV-2 the sticky part is the **spike protein**, and the target is **ACE2**, a protein on the outside of human cells. The stronger spike binds ACE2, the more easily the virus enters.



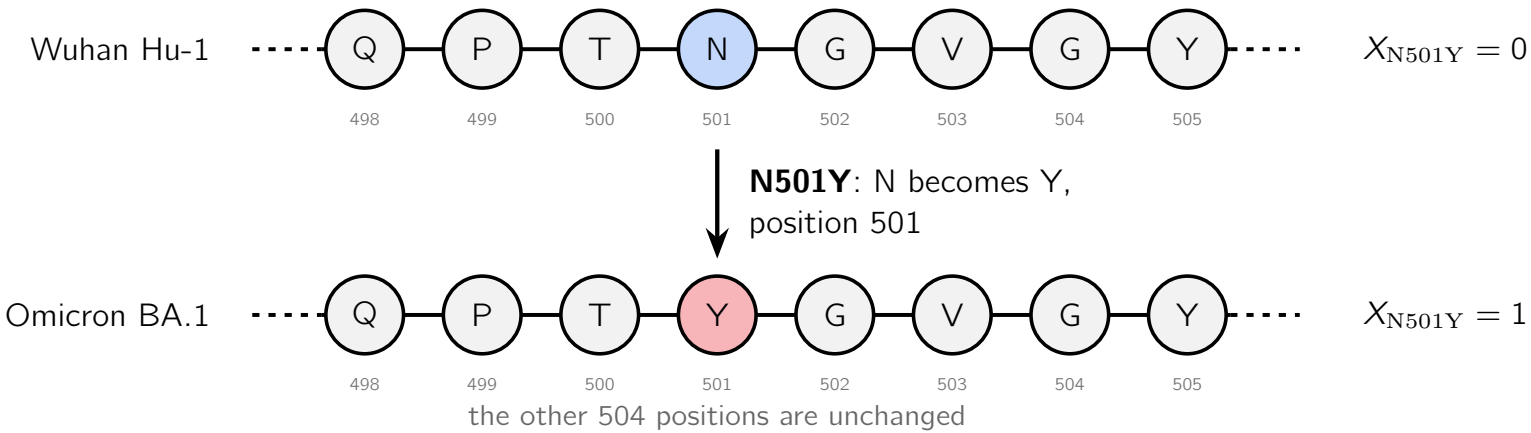
Background: modeling mutations

A protein is a chain of amino acids. A **mutation** swaps one amino acid for another at one position. The notation **N501Y** means position 501 changed from N (asparagine) to Y (tyrosine).

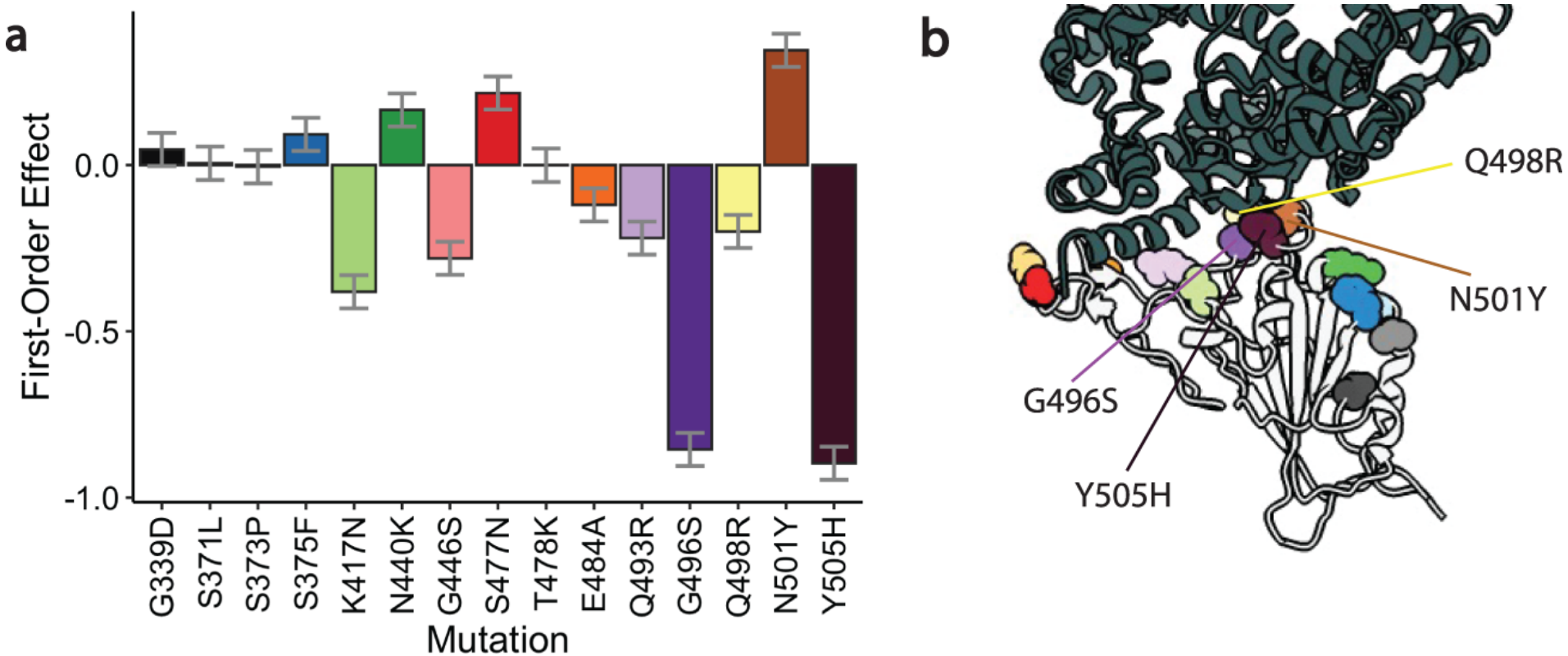


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Background: first order effects



(a) The effect of each mutation on binding, averaged over all backgrounds: N501Y helps, Y505H and G496S hurt. (b) The spike (white) against ACE2 (dark), with the mutated positions as colored blobs.

Figures from Moulana et al. (2022), *Nat. Commun.*

Fitted model

A variant is described by 15 indicators X_i where $X_i = 1$ if mutation i is present, $X_i = 0$ if the position still carries the original amino acid. For example, $X_{\mathrm{Y505H}} = 1$ means position 505 has the Omicron amino acid at that site.

data: all 2^{15} variants

X_{Q498R}	X_{N501Y}	X_{Y505H}	X_{K417N}	$\cdots$	Y
0	0	0	0	$\cdots$	9.03
0	0	1	0	$\cdots$	7.39
0	0	1	1	$\cdots$	7.94
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$
1	1	1	1	$\cdots$	9.48

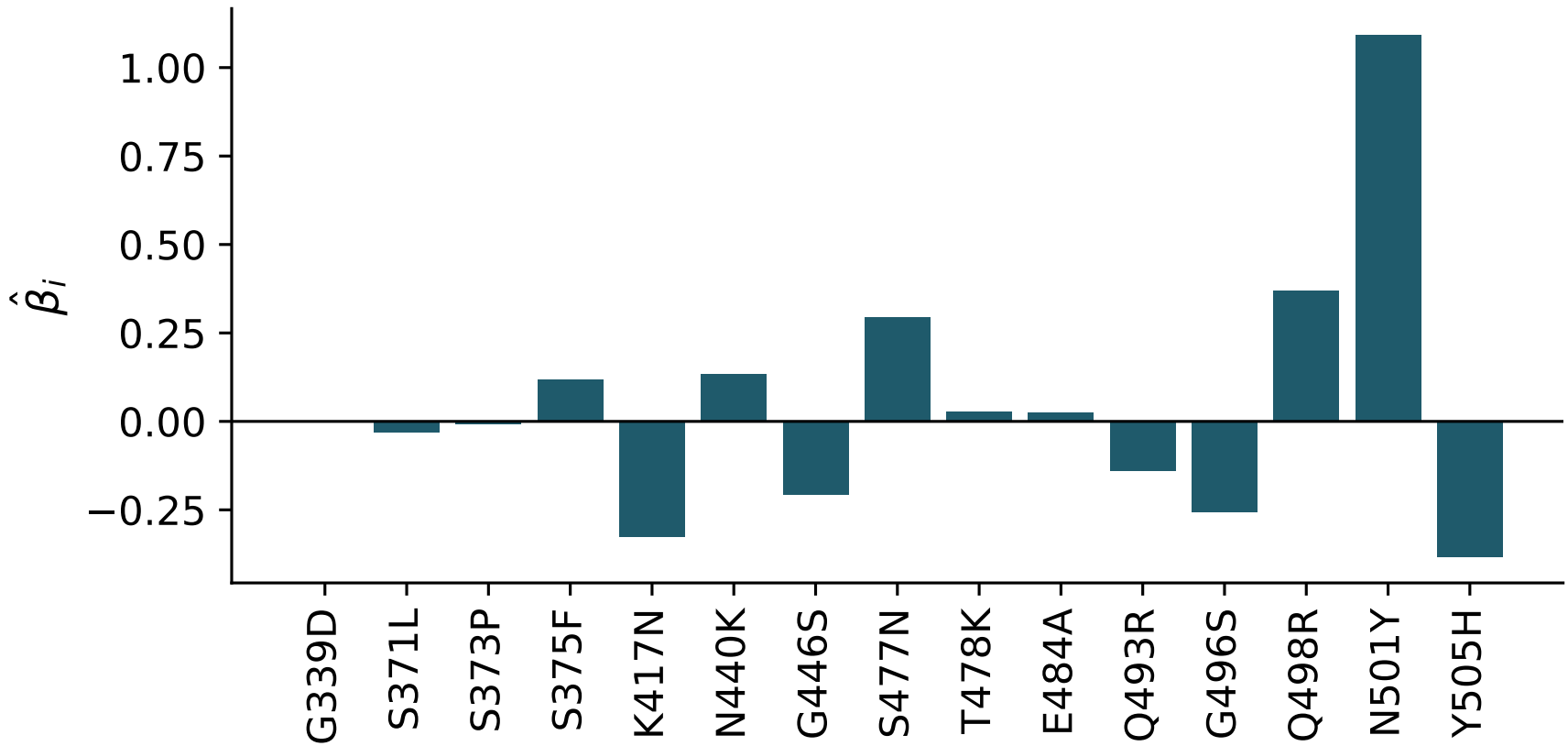
$\cdots$ are the other 11 mutations

probability model

$$Y = \beta_0 + \sum_{i=1}^{15} \beta_i X_i + \epsilon$$

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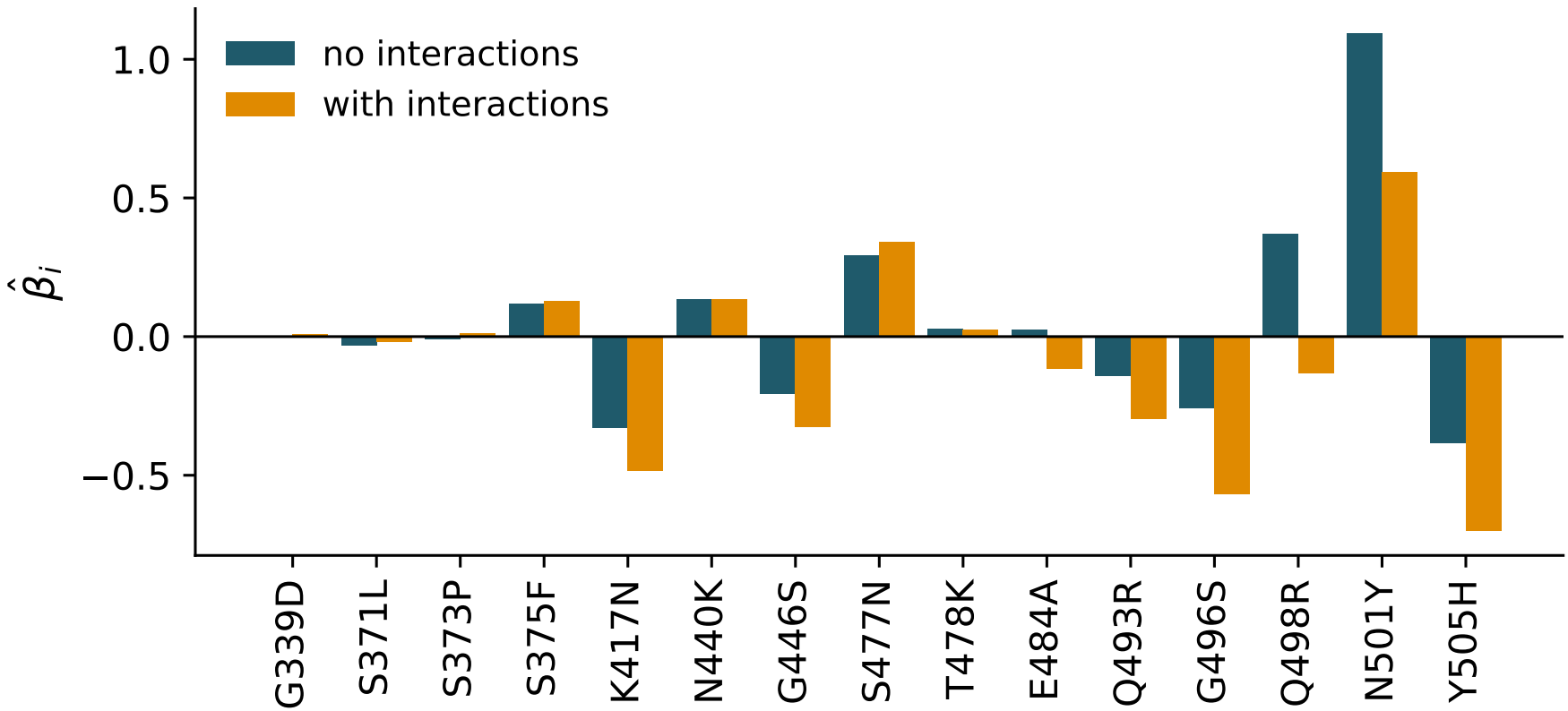
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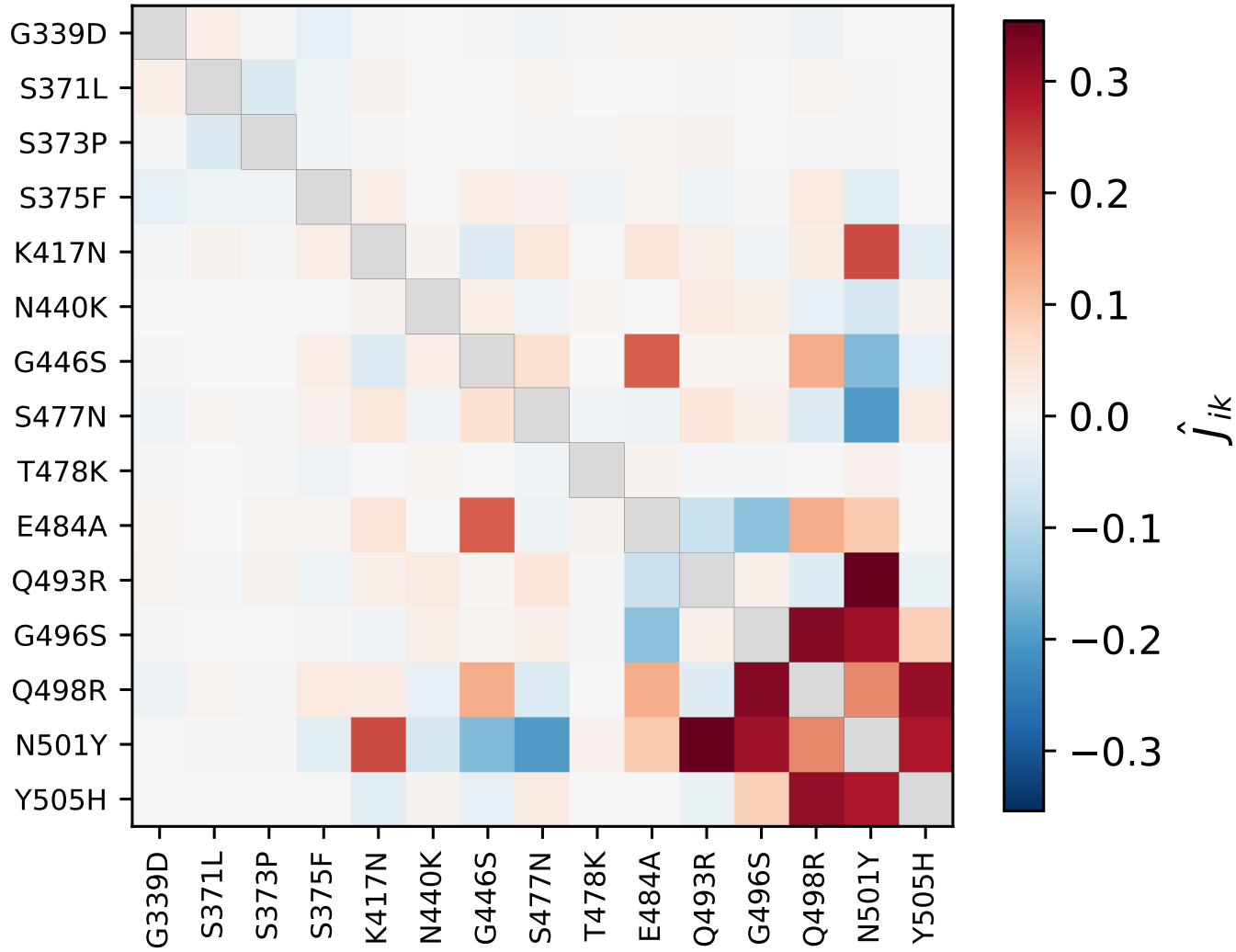
probability model

$$Y = \beta_0 + \sum_{i=1}^{15} \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \epsilon$$

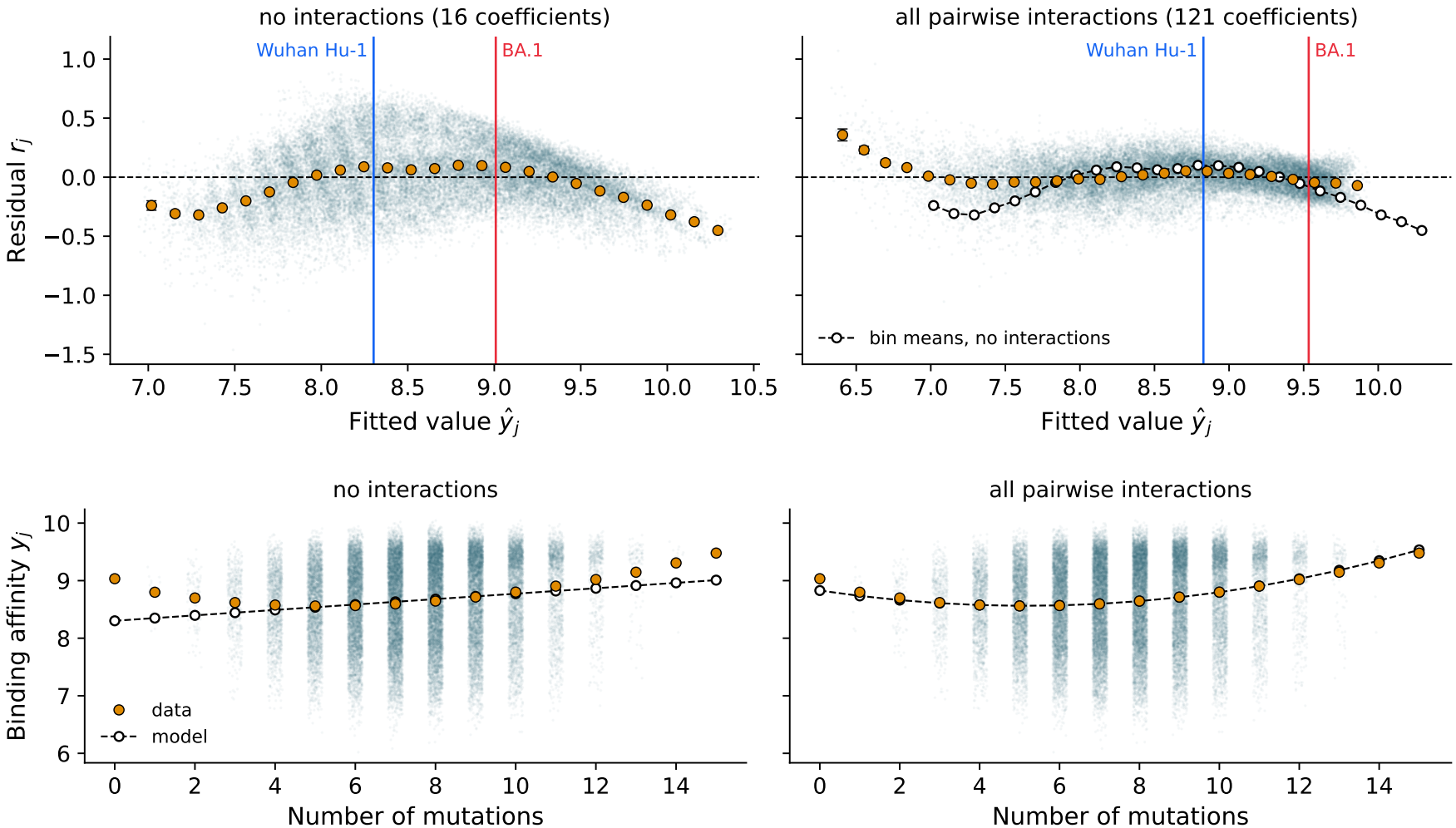
First-order coefficients: both models



Interaction coefficients



Residuals: both models



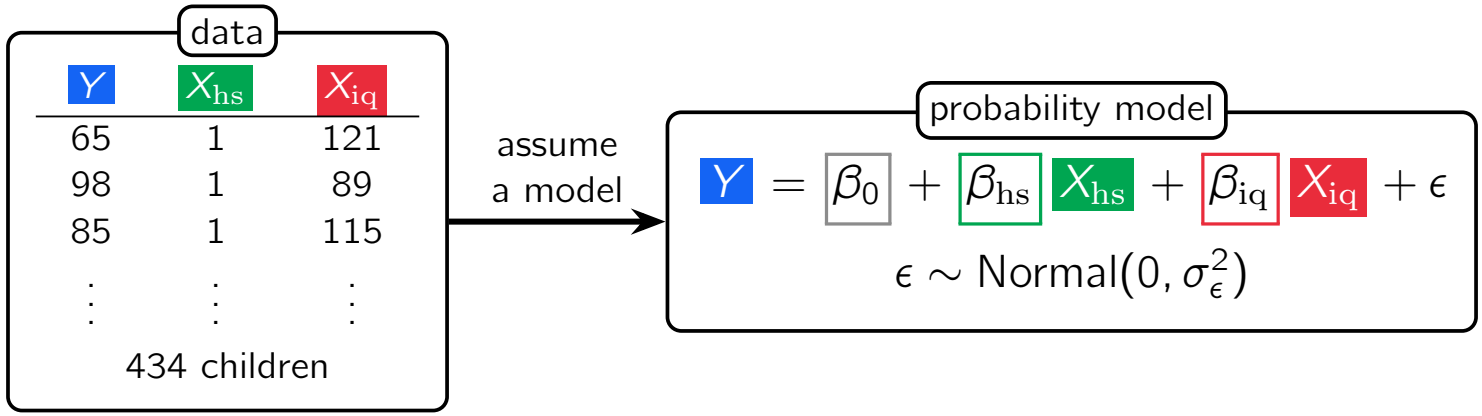
Statstics: Recap

data

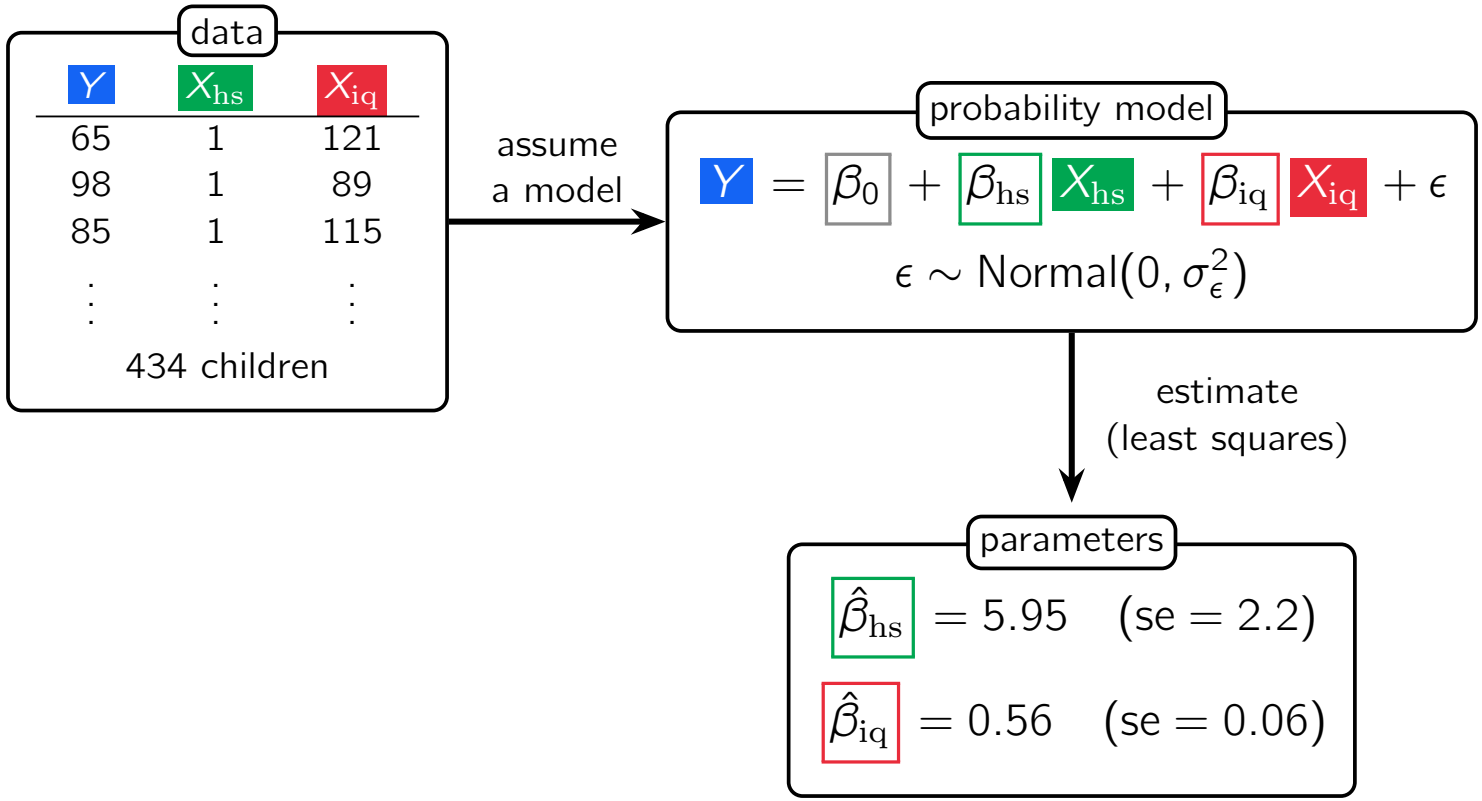
Y	X_{hs}	X_{iq}
65	1	121
98	1	89
85	1	115
⋮	⋮	⋮

434 children

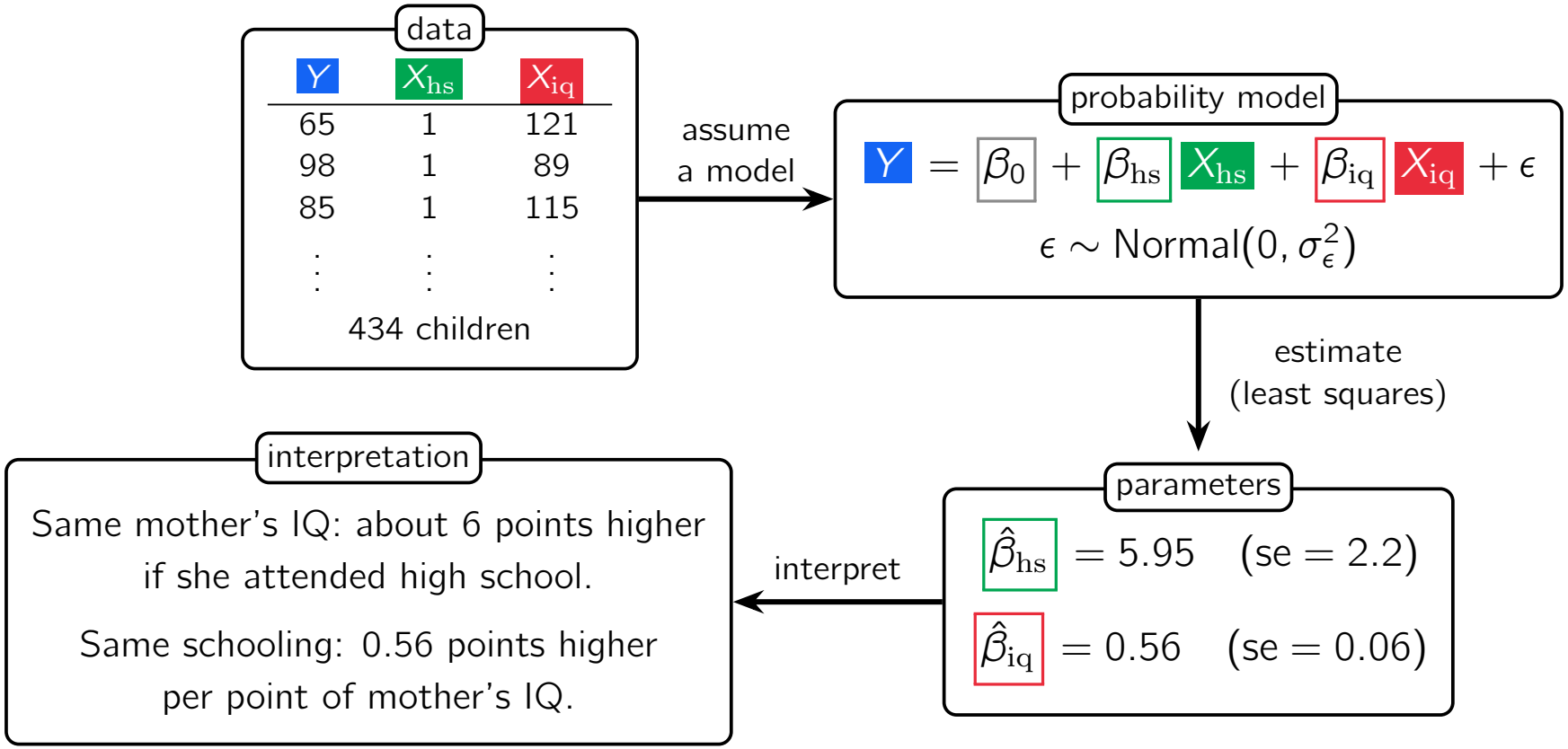
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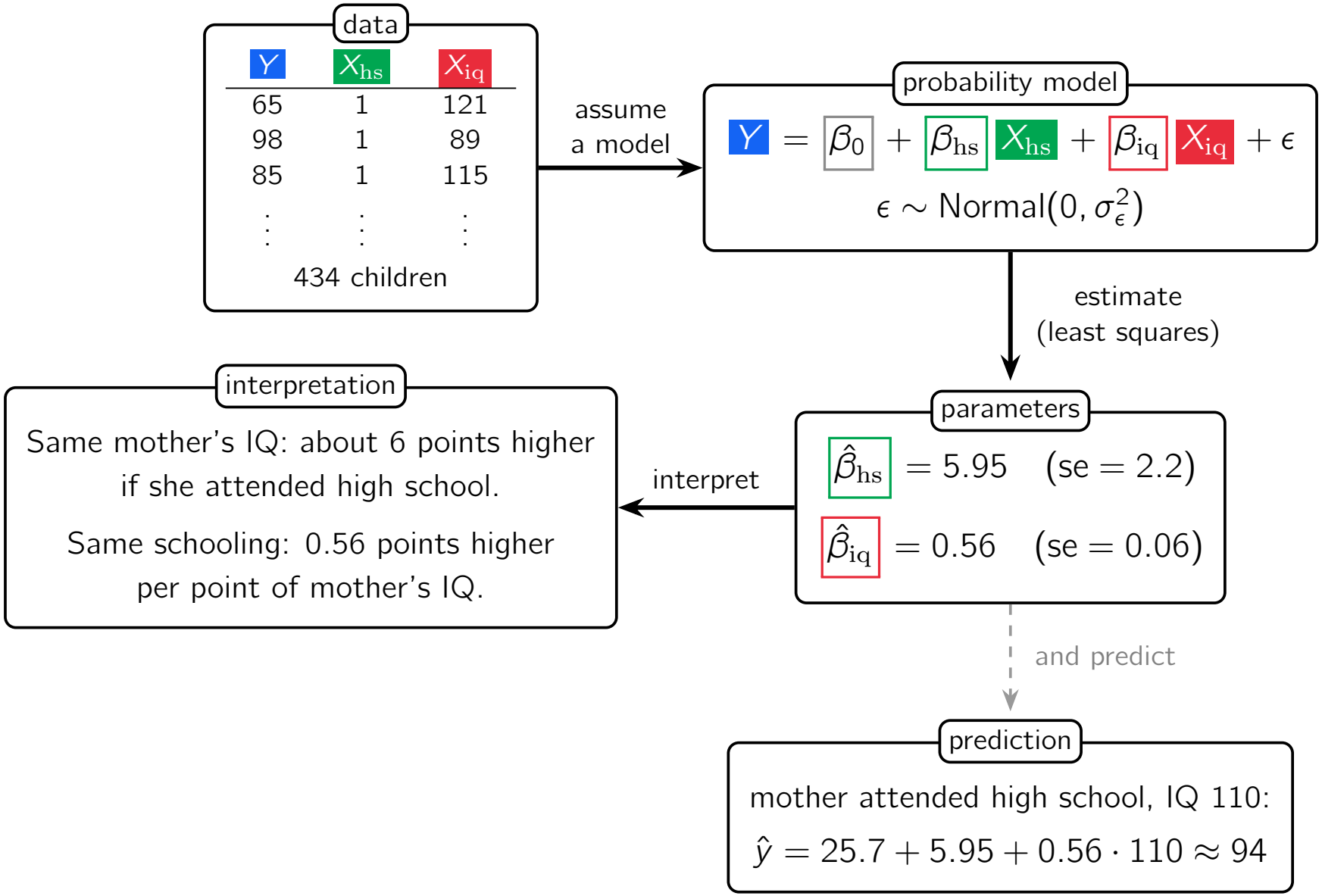
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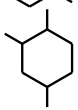


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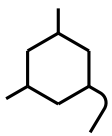


Example: drug discovery

data: N molecules

molecule	binding energy
	y_1
	y_2
	y_3
⋮	⋮
	y_N

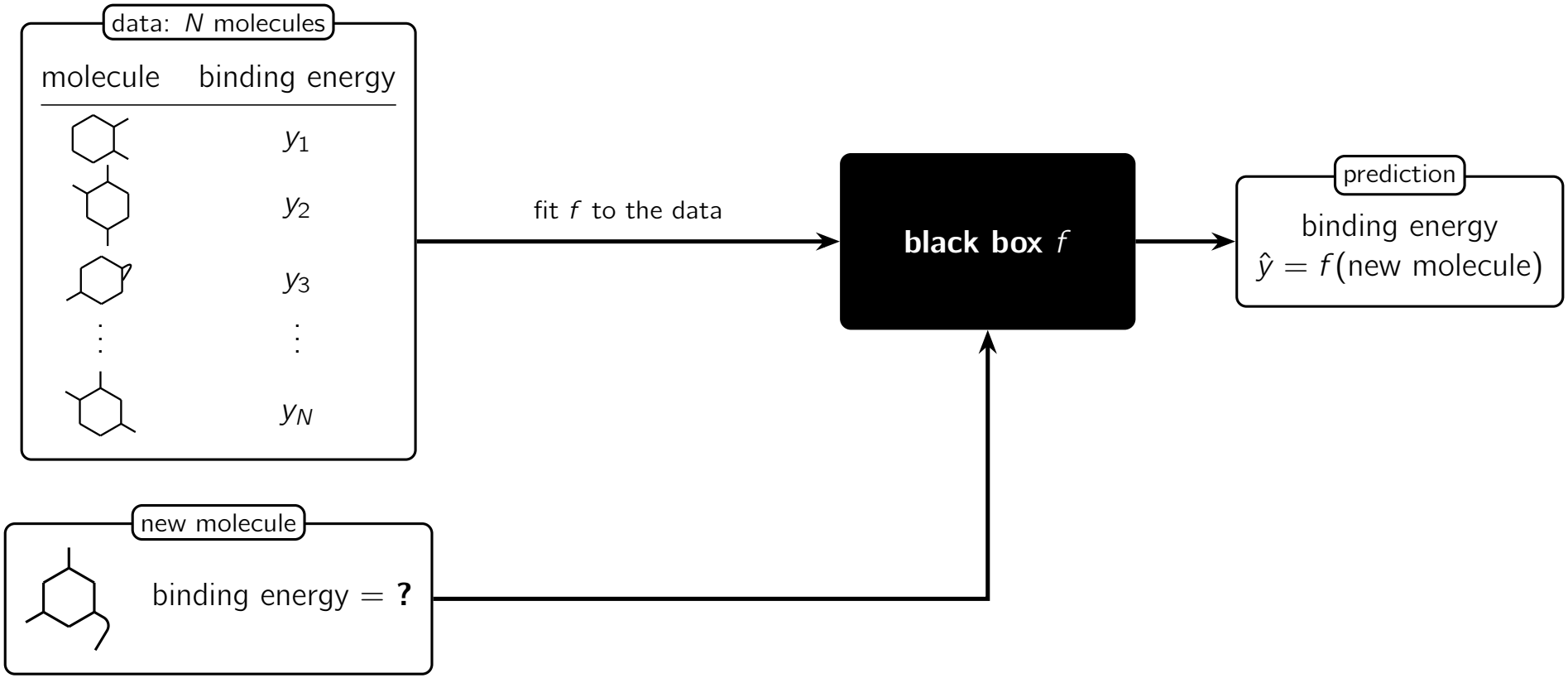
new molecule



binding energy = **?**

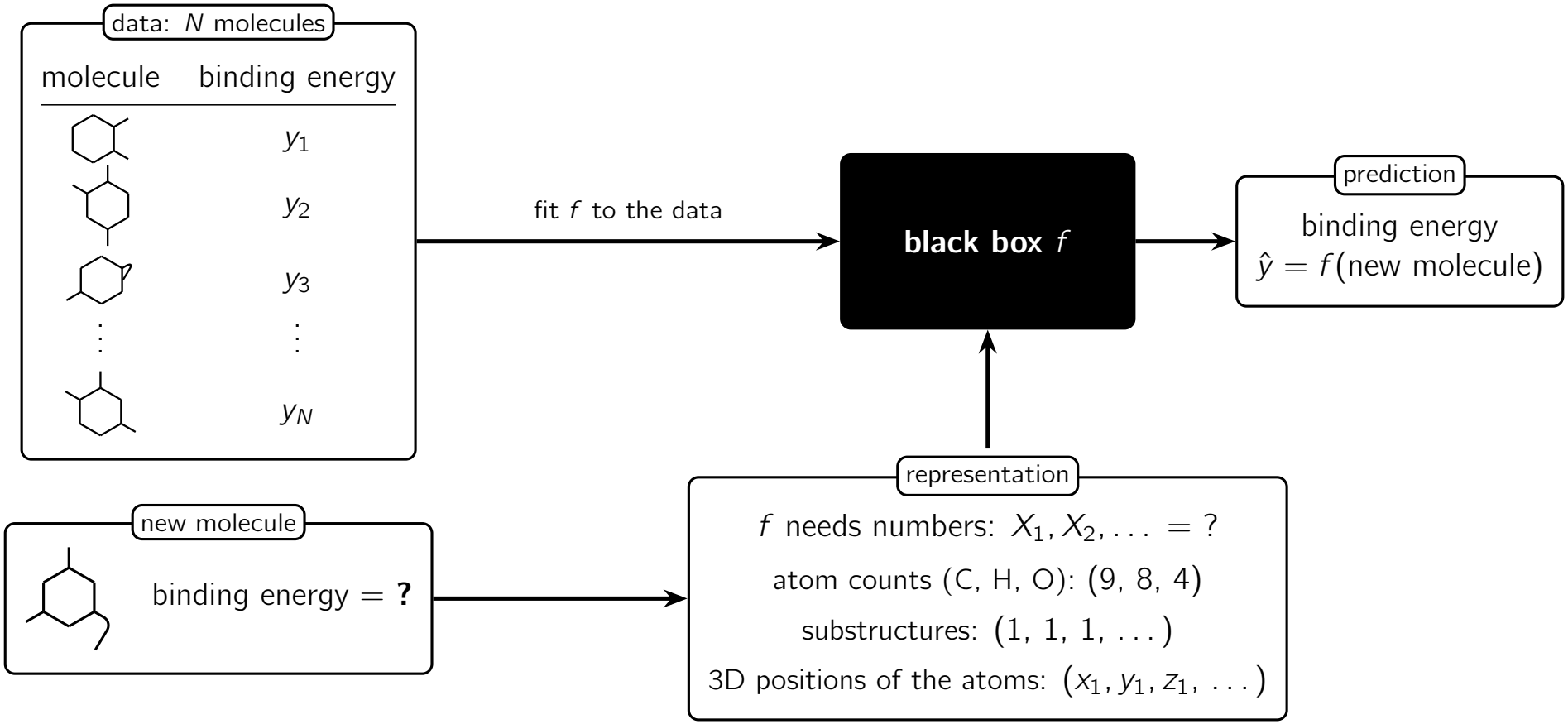
The input is a chemical structure which can be represented in different ways. The more complex input data and choice of representation makes this very different from the test scores and other examples.

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Example: a language model

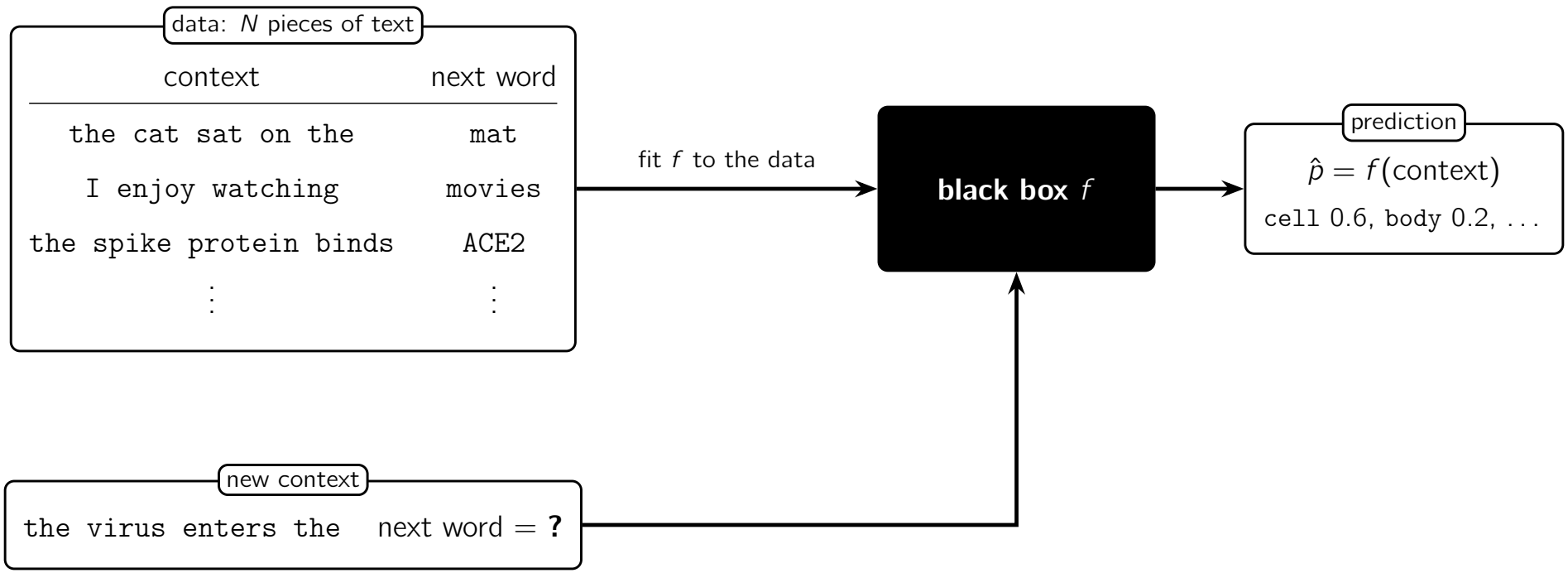
data: N pieces of text

context	next word
`the cat sat on the`	`mat`
`I enjoy watching`	`movies`
`the spike protein binds`	`ACE2`
⋮	⋮

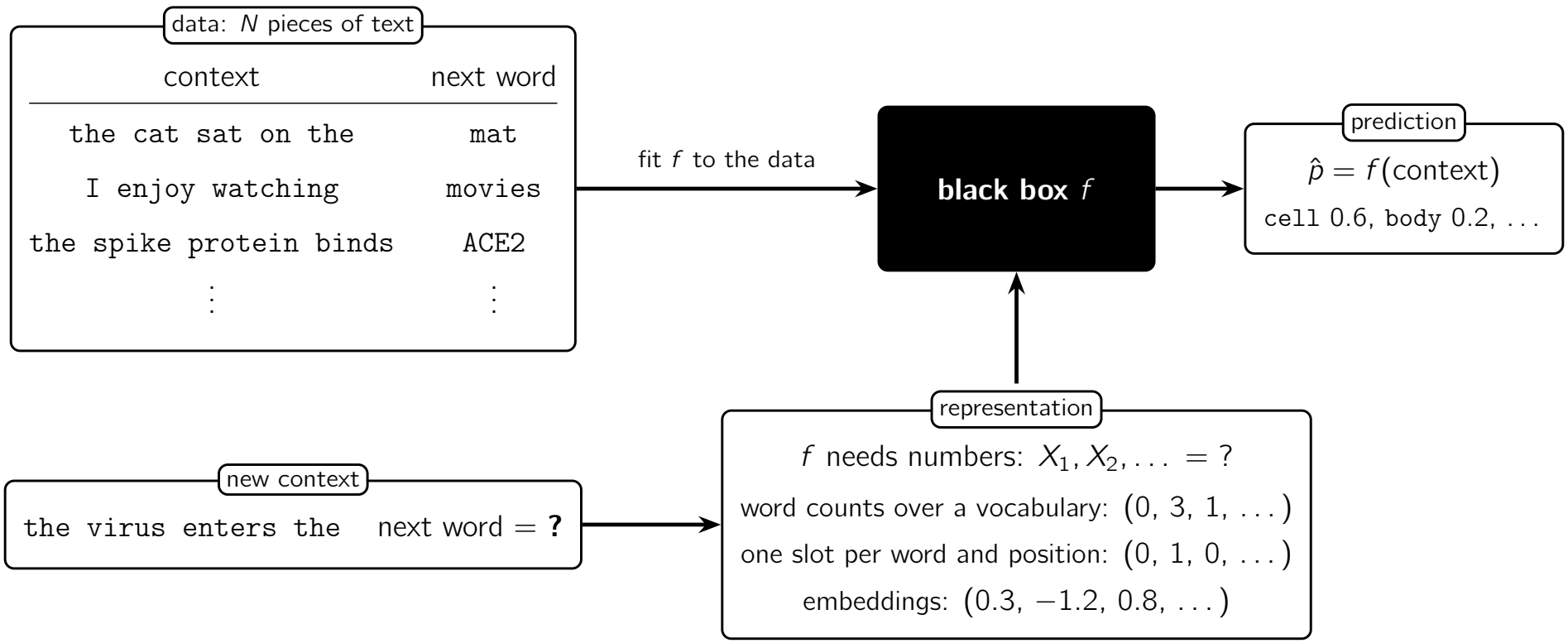
new context

`the virus enters the` next word = **?**

Example: a language model



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What is the difference between statistics and ML?

Proposed answer	Objection

Further reading: L. Breiman (2001), Statistical modeling: the two cultures, *Statistical Science* 16(3), 199–231.

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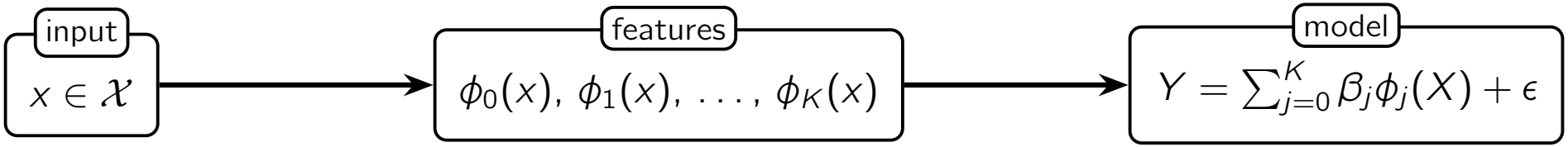
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A theme is complexity. What does it mean for a model to be complex? Does having more parameters make it more complex? Can a model be too complex?

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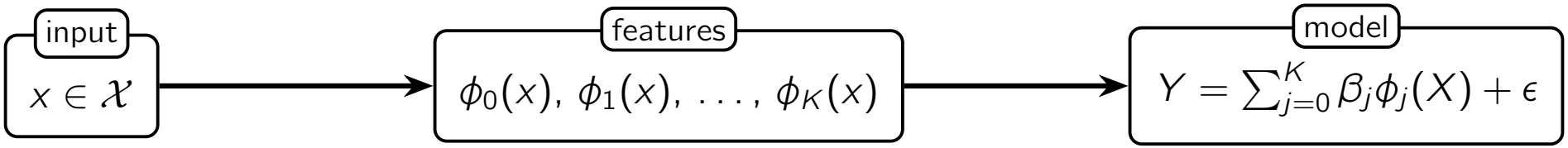
Learning a function

There is an unknown function f from a space of inputs $\mathcal{X}$ to $\mathbb{R}$, and we observe $Y = f(X) + \epsilon$. We want to learn what f looks like.



Learning a function

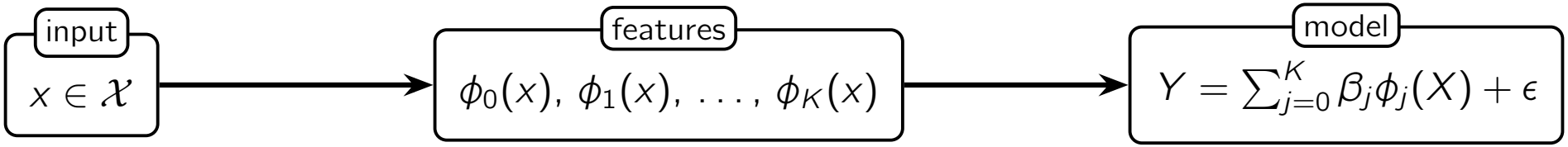
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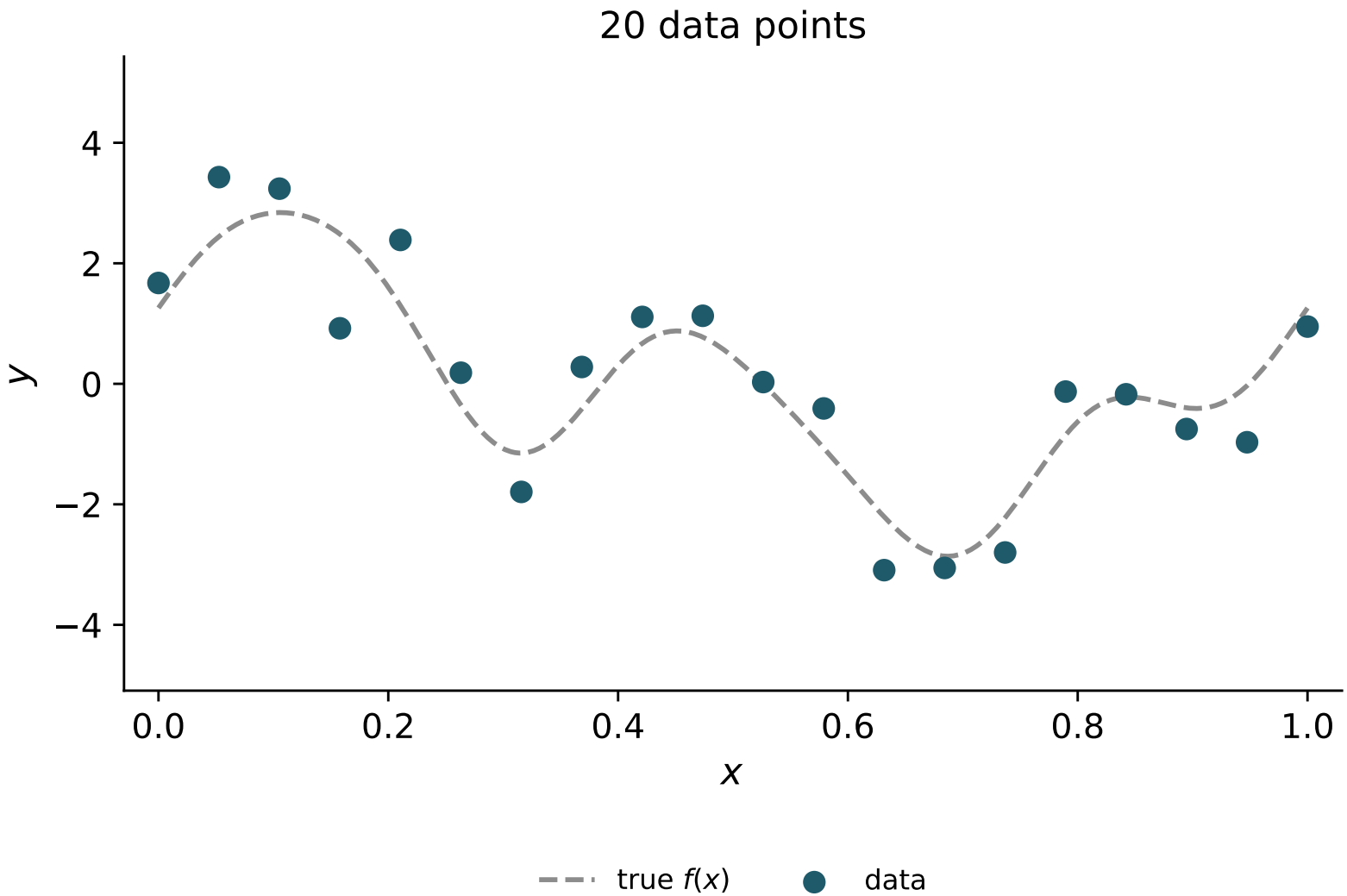


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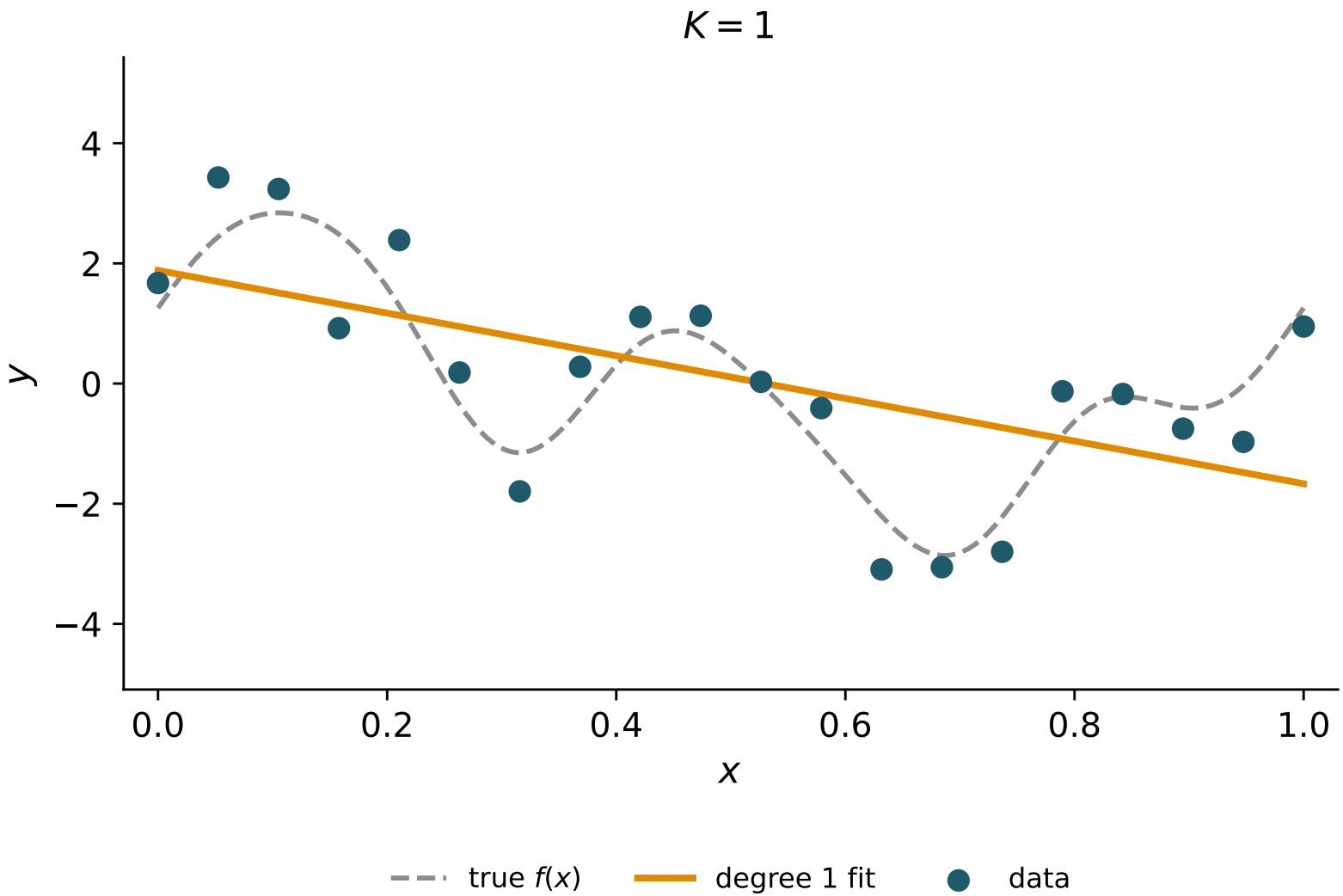
Polynomial regression: $\mathcal{X} = \mathbb{R}$ and $\phi_j(x) = x^j$, so

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_K X^K + \epsilon.$$

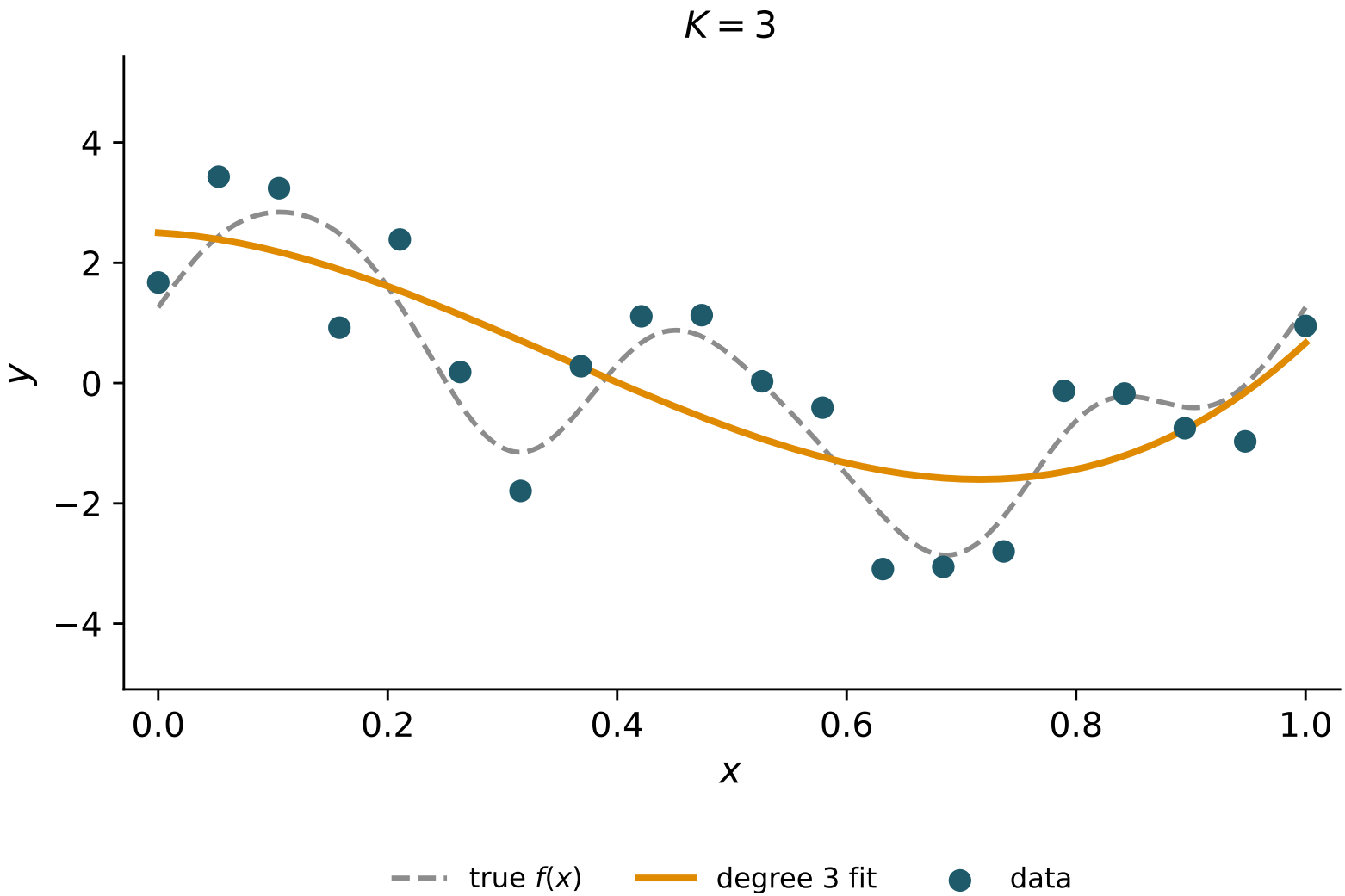
Example: polynomials fitted to a random function



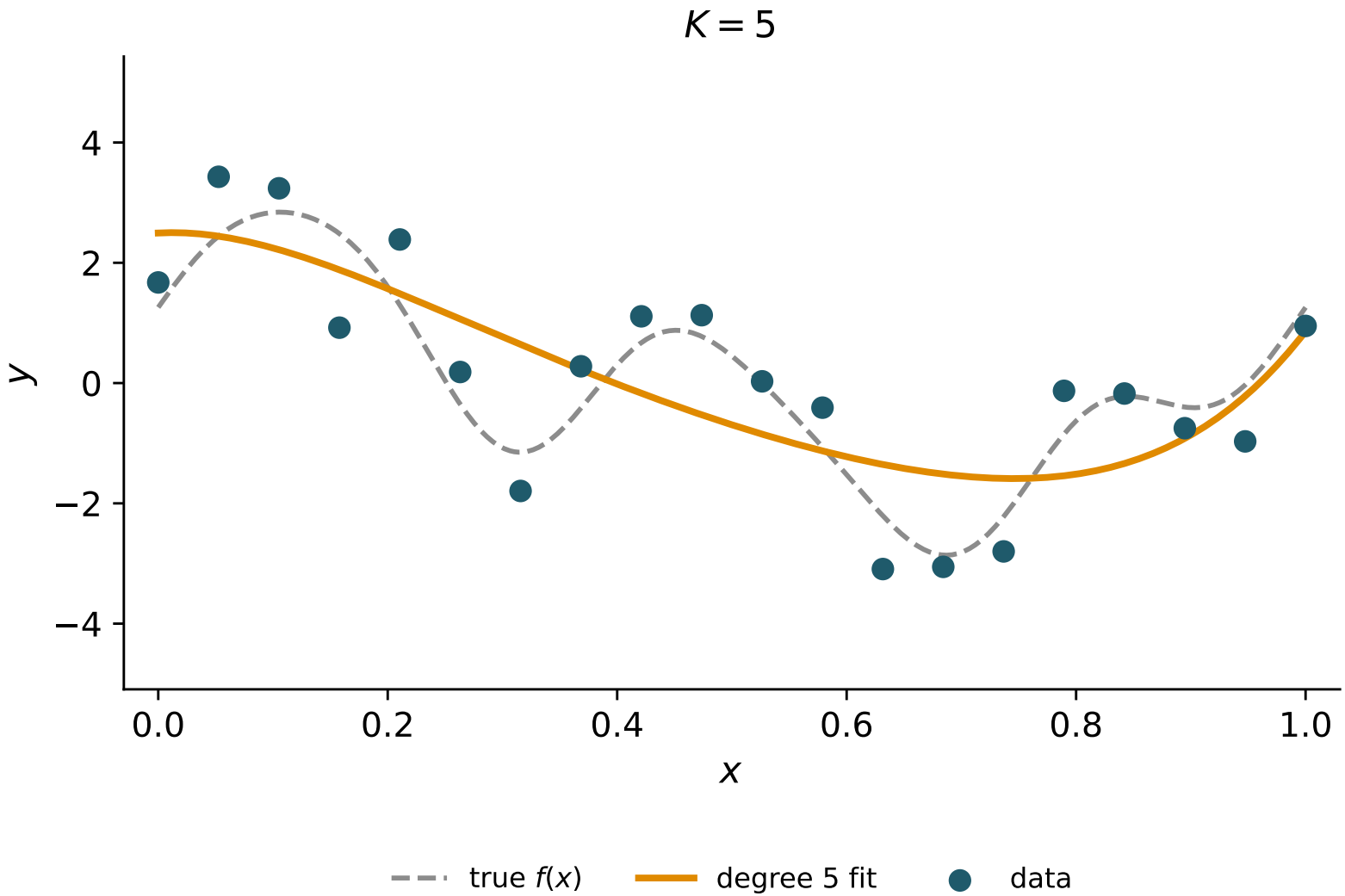
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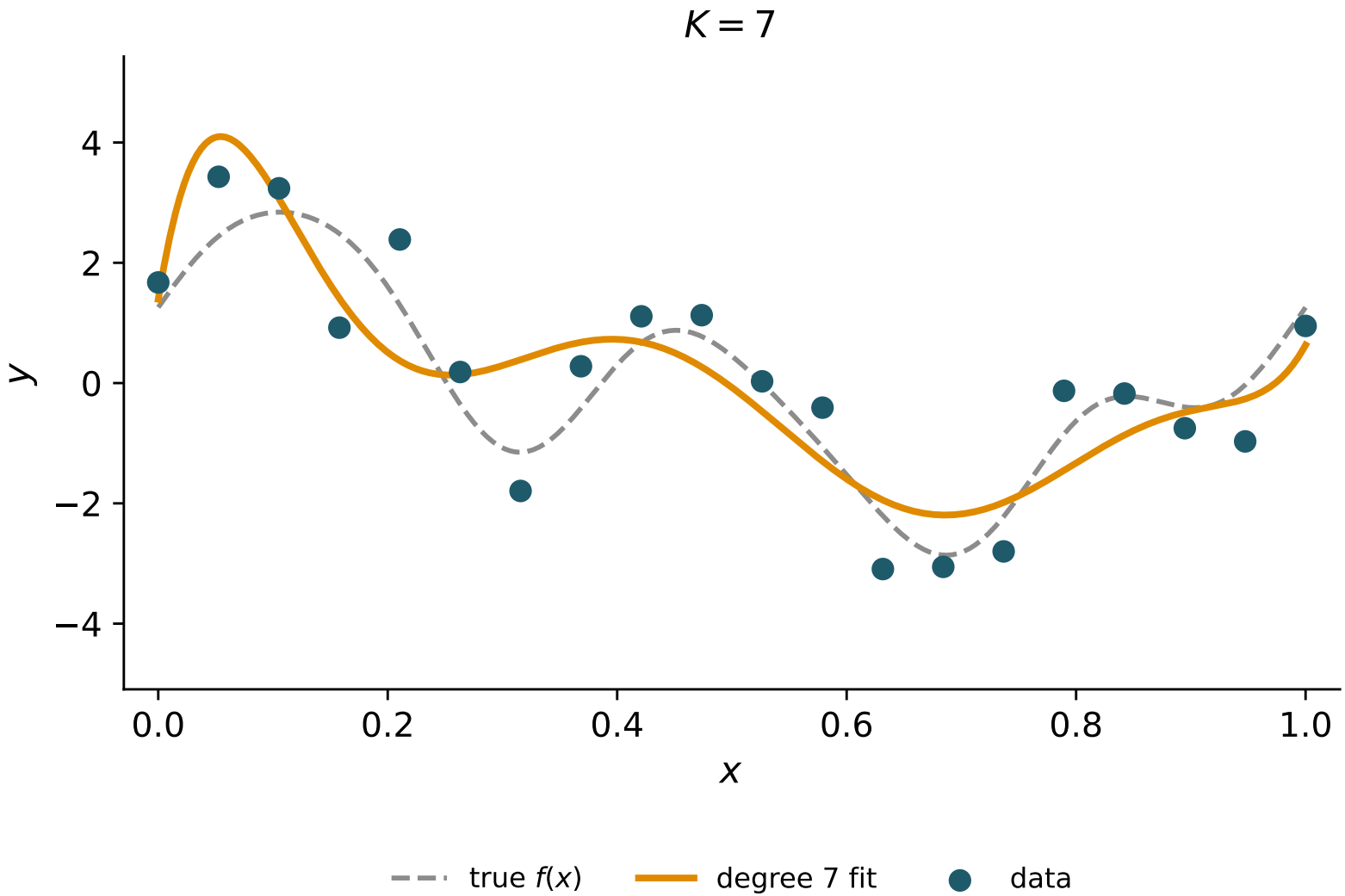
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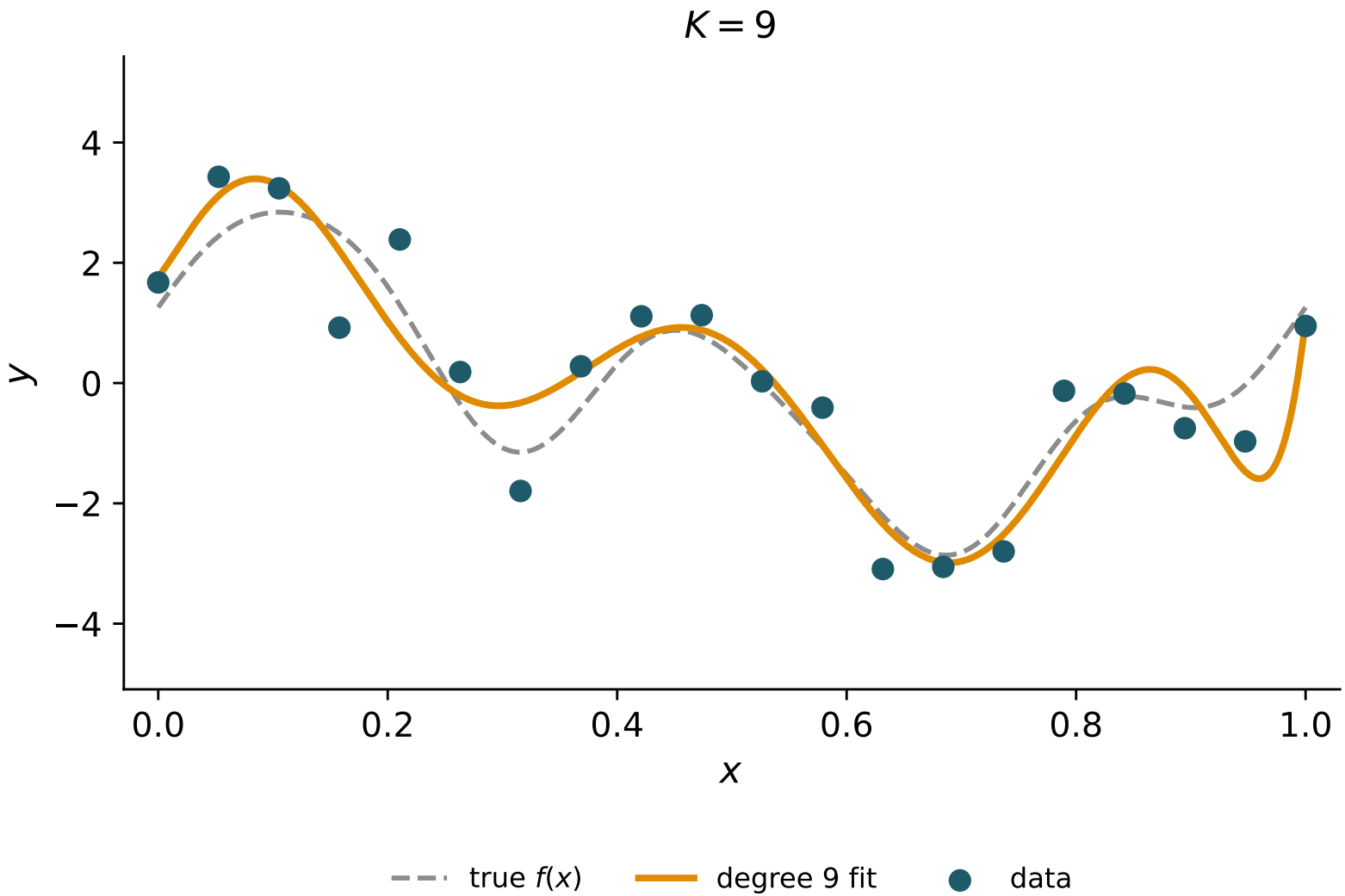
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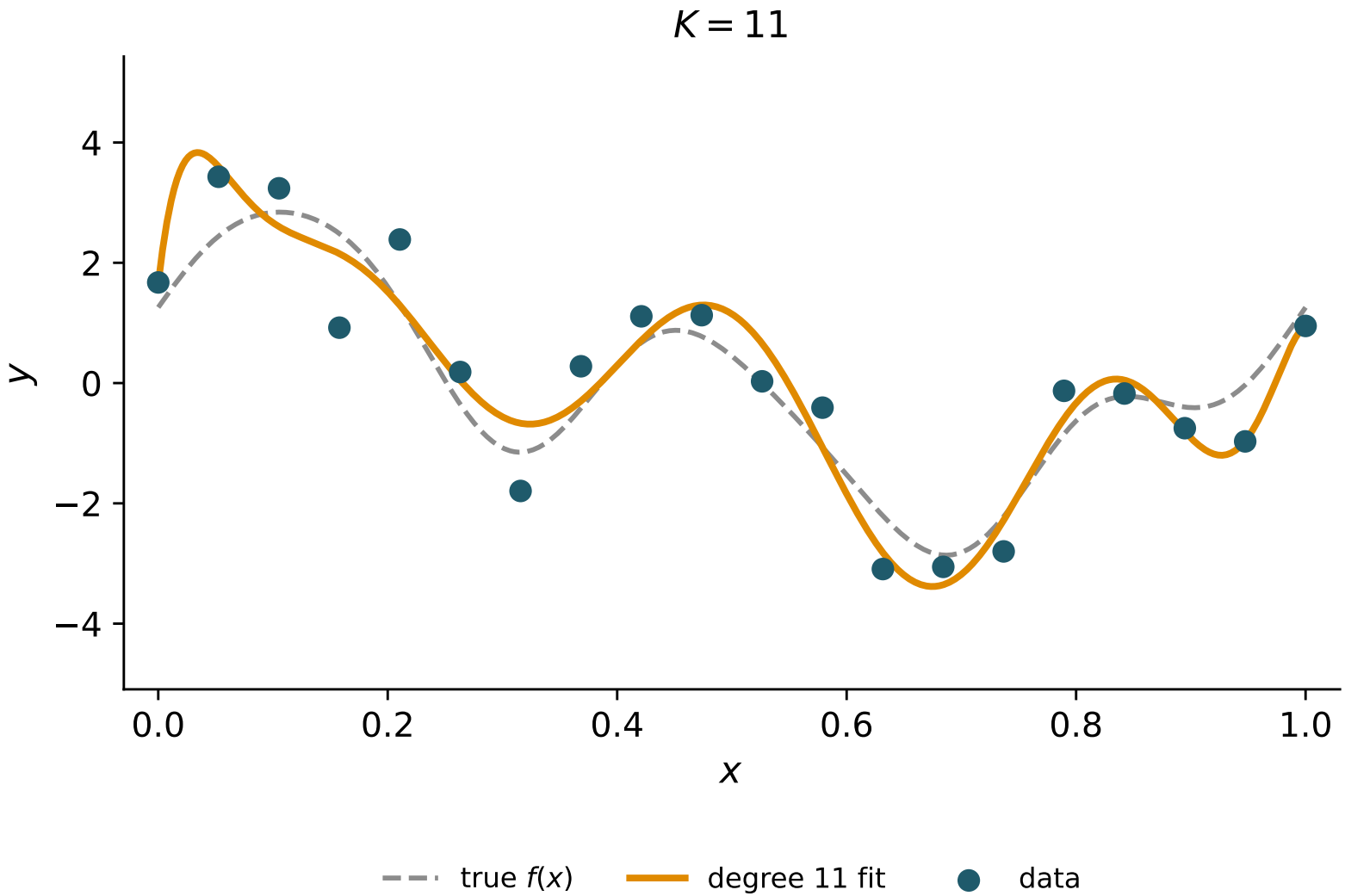
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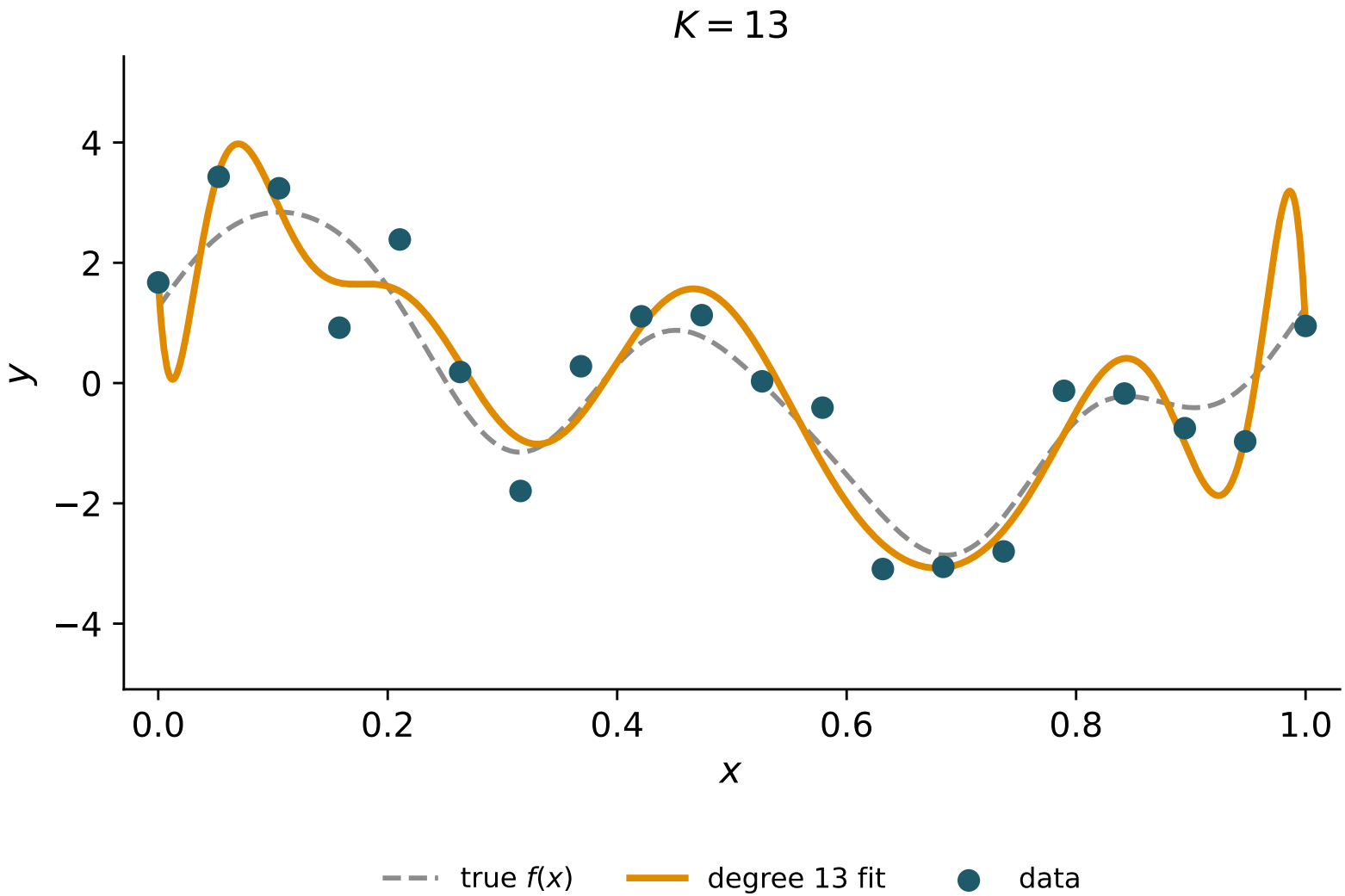
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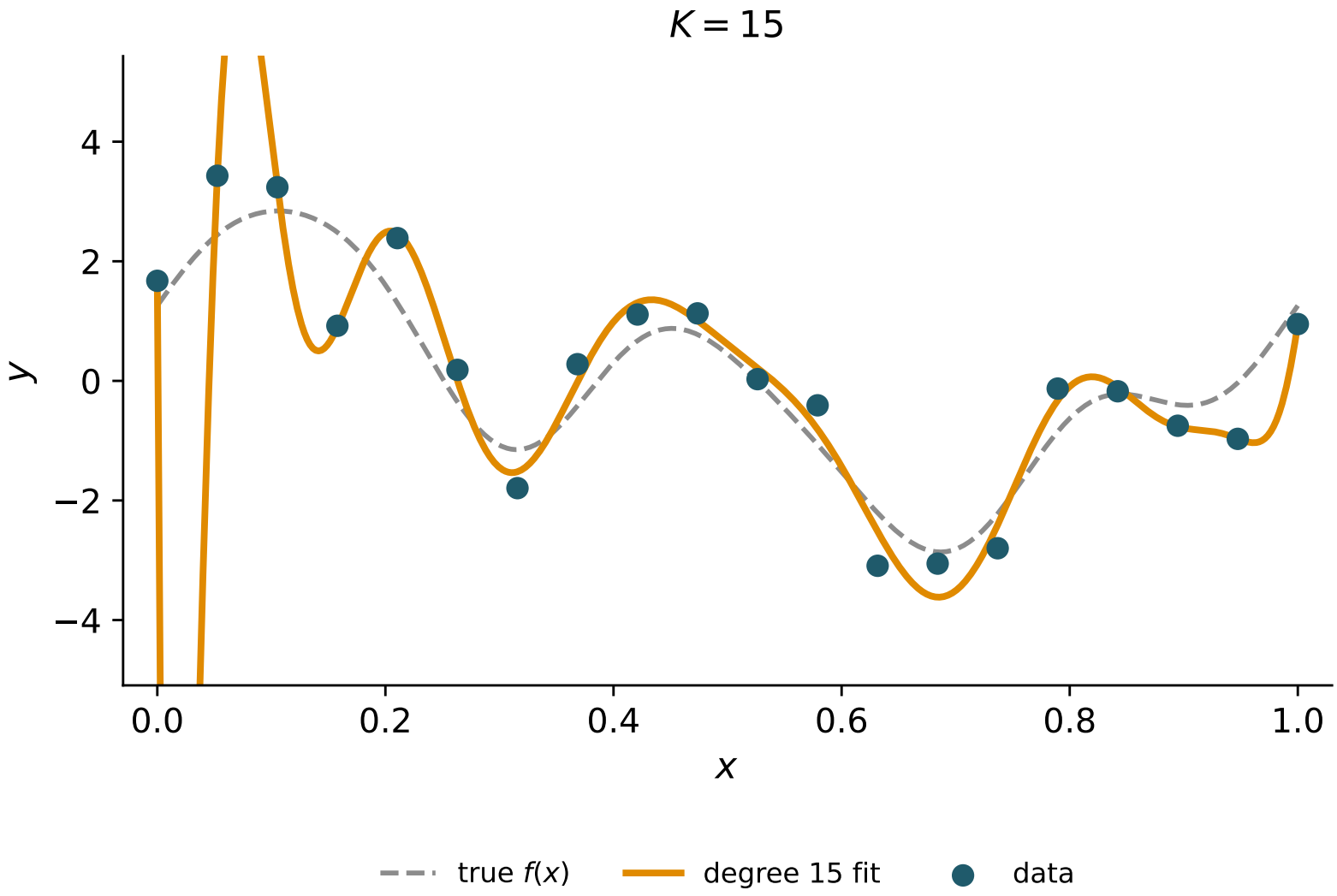
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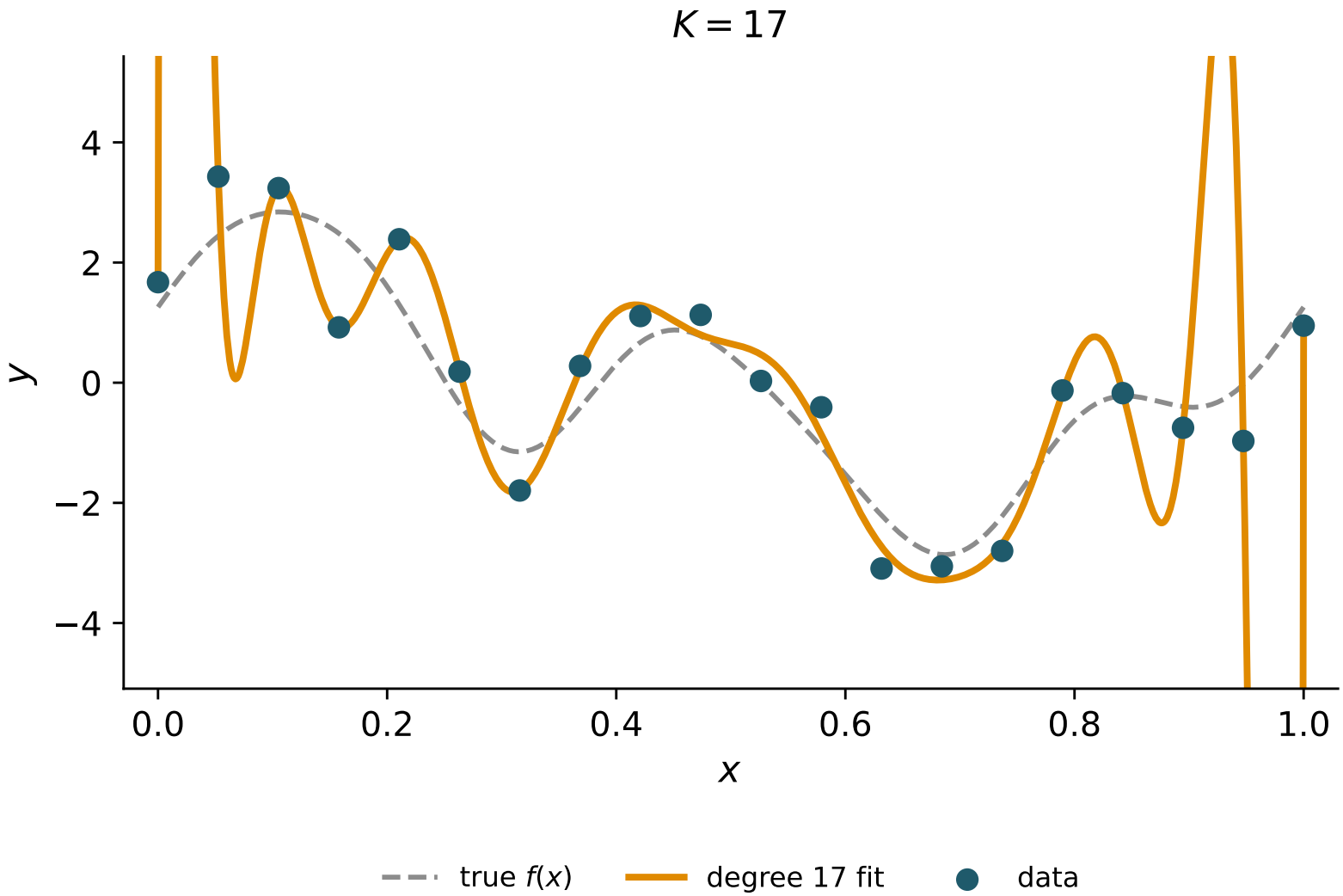
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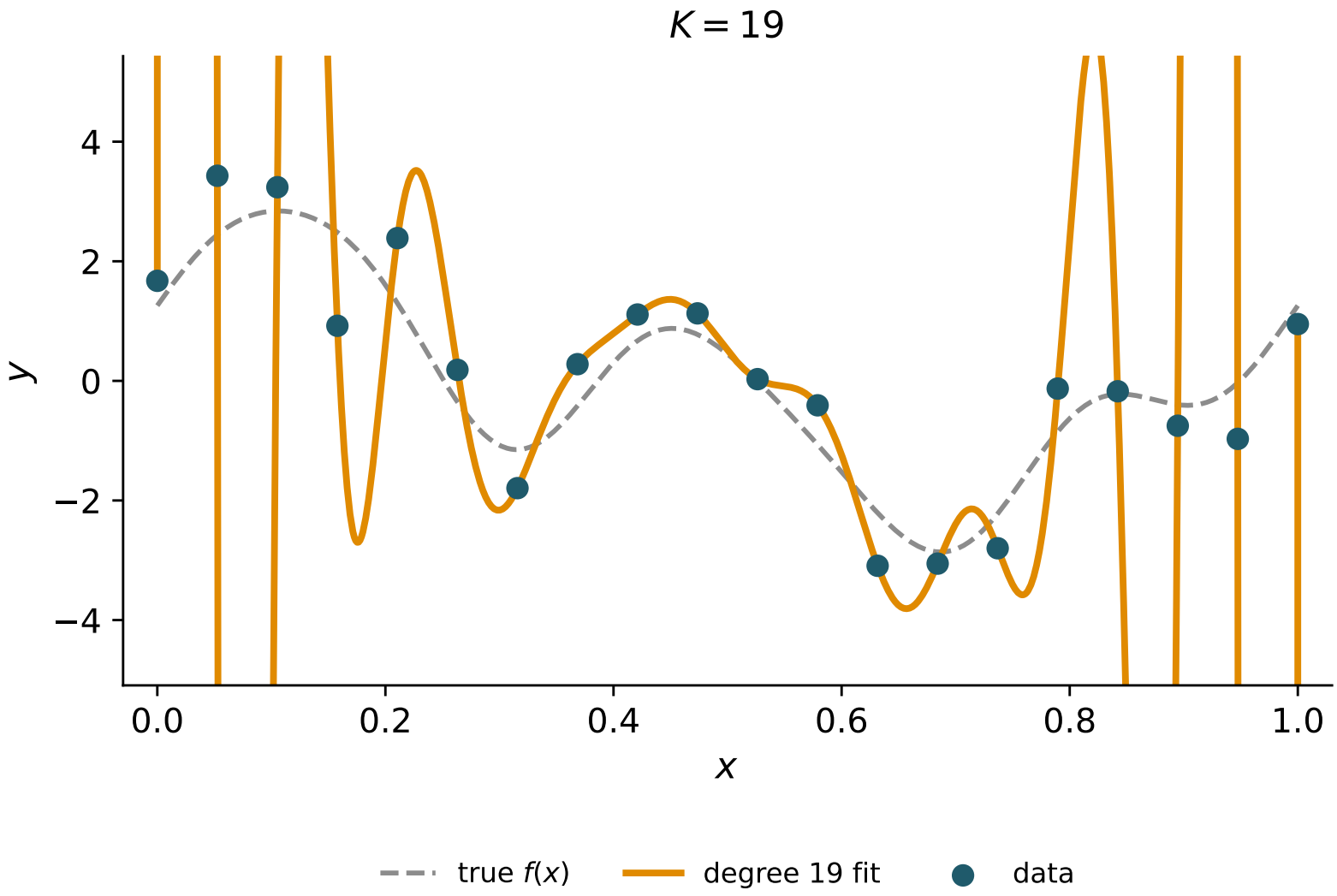
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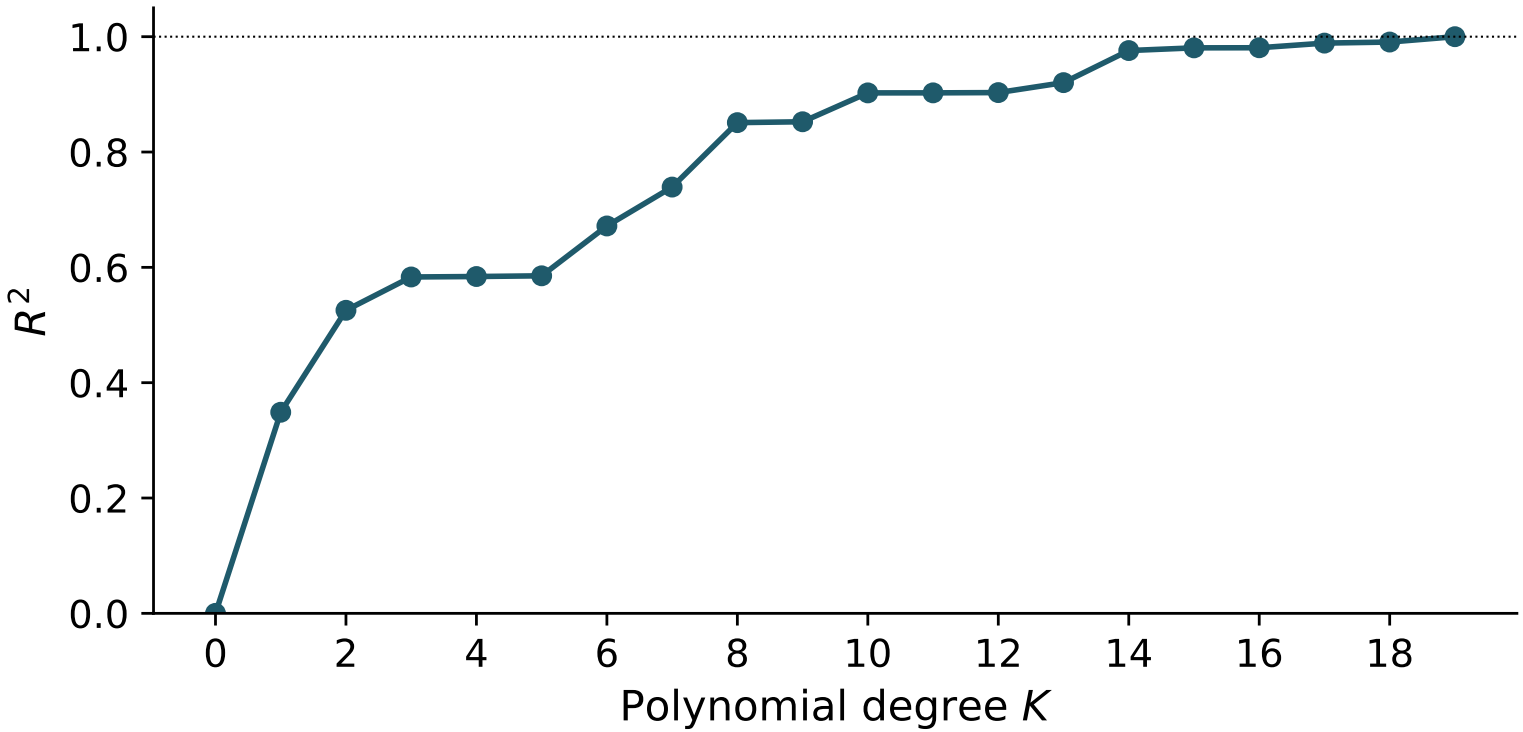


Example: polynomials fitted to a random function



R^2 as the degree increases

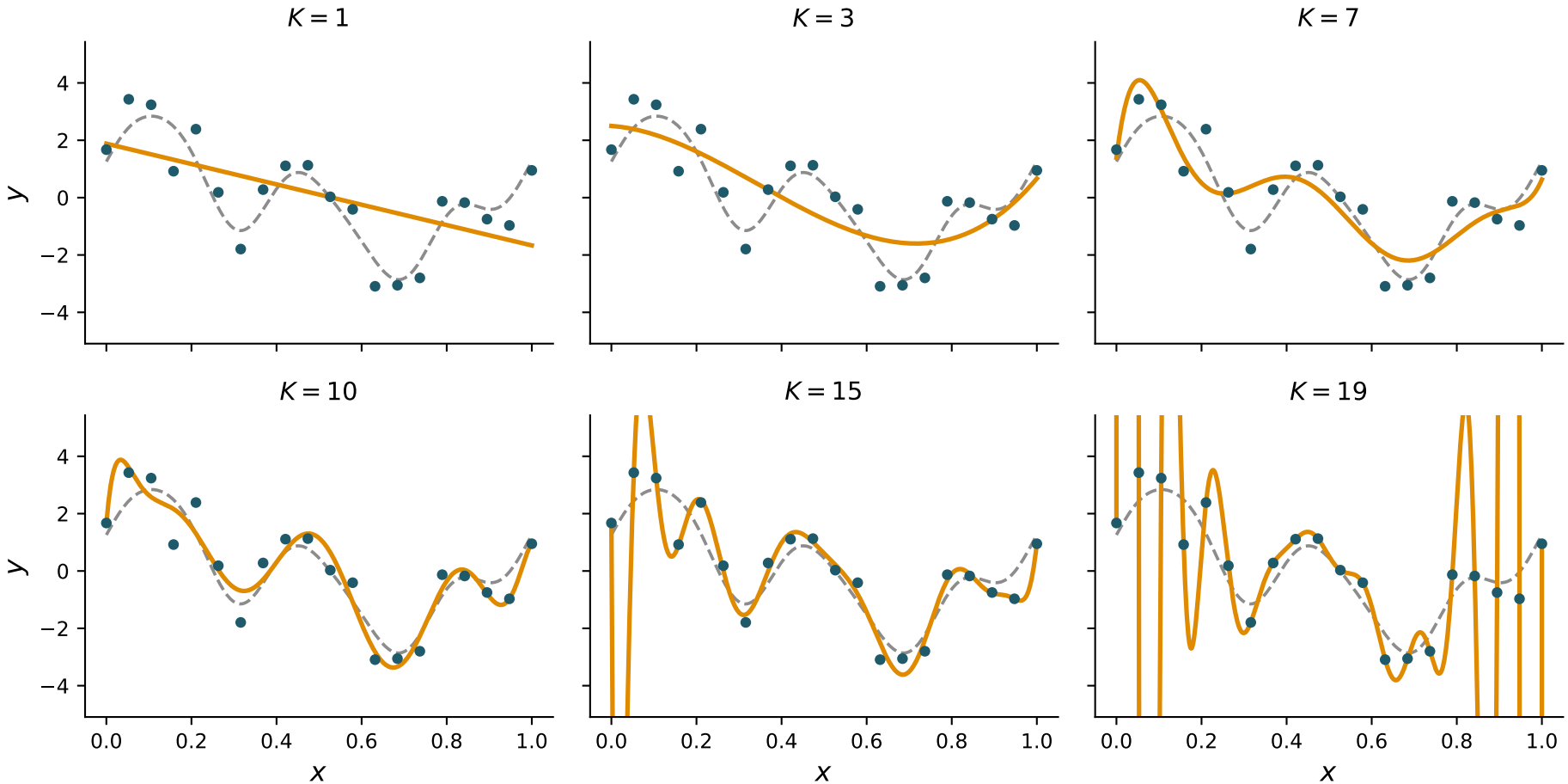
R^2 for the same 20 points and every degree $K = 0, 1, \ldots, 19$:



R^2 goes up with every coefficient we add and reaches 1 at $K = 19$, where the polynomial passes through all 20 points.

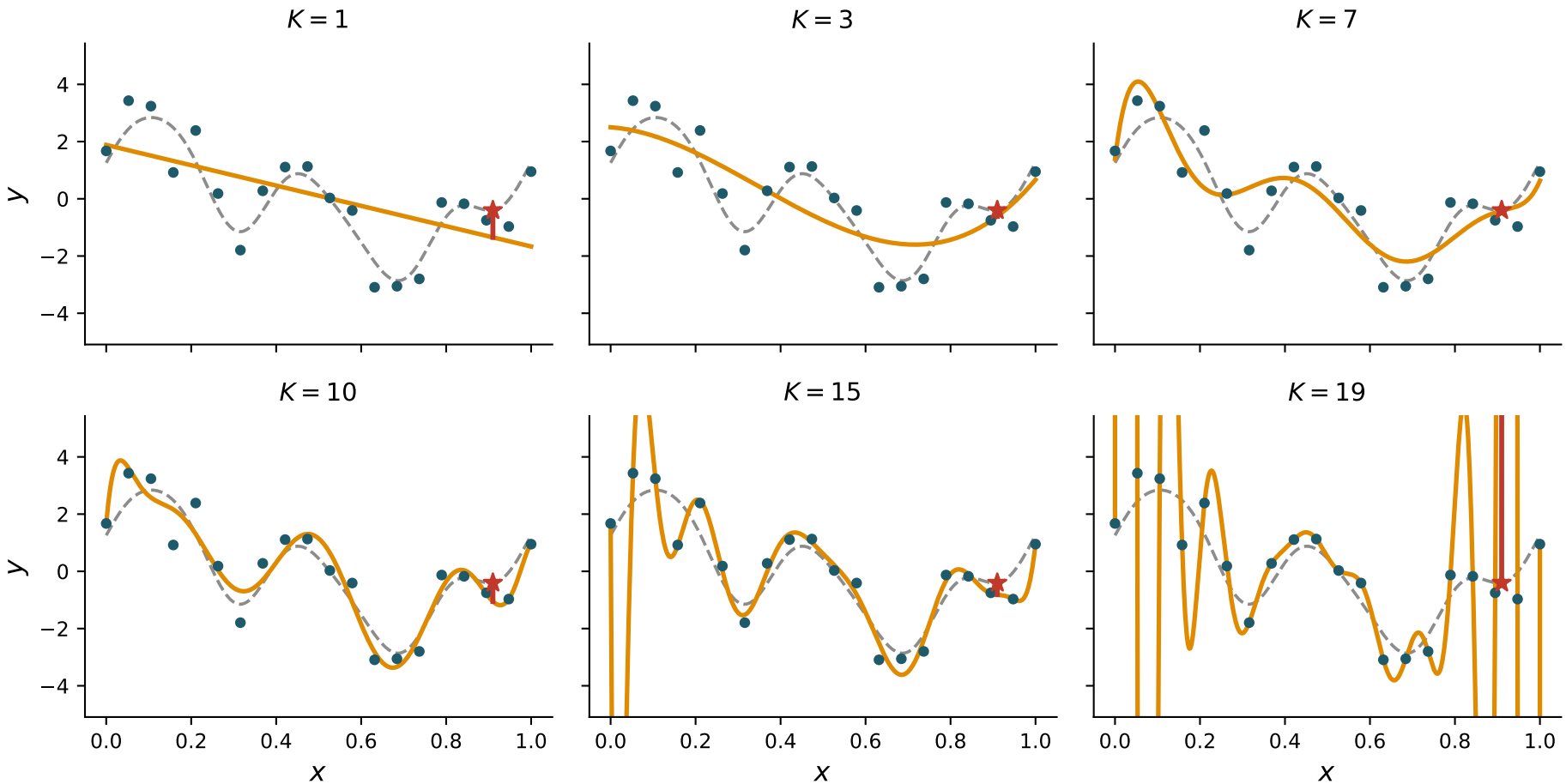
Predicting a new point

The true function (dashed), the 20 points and the fitted polynomials.



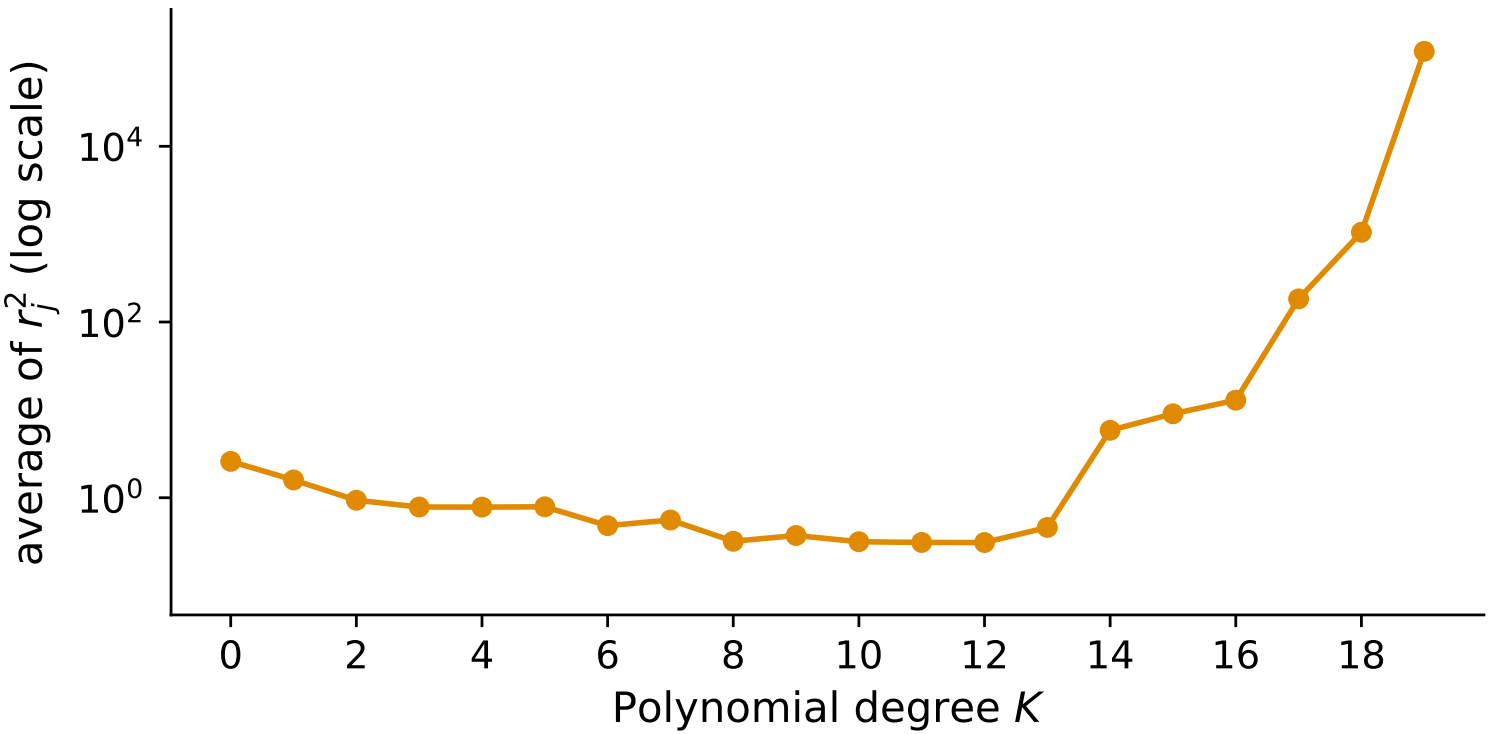
Predicting a new point

The true function (dashed), the 20 points and the fitted polynomials. A new point on the true function at $x = 0.91$ (star). Its residual $r = y - \hat{y}$ is the vertical distance from the fitted curve (red).



Average squared residual of the new points

Repeat this for 200 random points on the true function, with residuals r_j. For each degree K, the average of r_j^2:



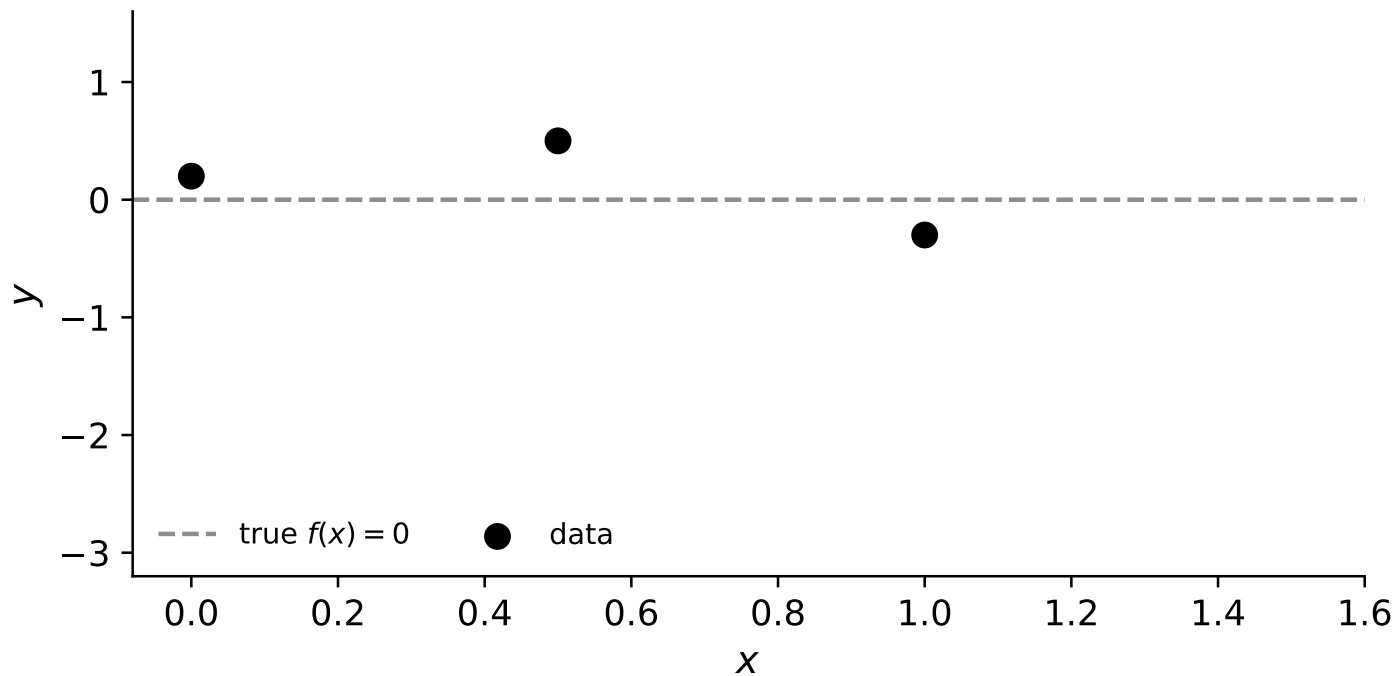
It is smallest around $K = 10$ to 12. For larger K it grows quickly, and at $K = 19$ it is about 120000.

Why the coefficients blow-up: minimal example

Consider the model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^3 + \epsilon$$

Assume the data is generated with $\beta_0 = \beta_1 = \beta_2 = 0$, so there is no underlying relationship. Three data points, at $x = 0$, 0.5 and 1.

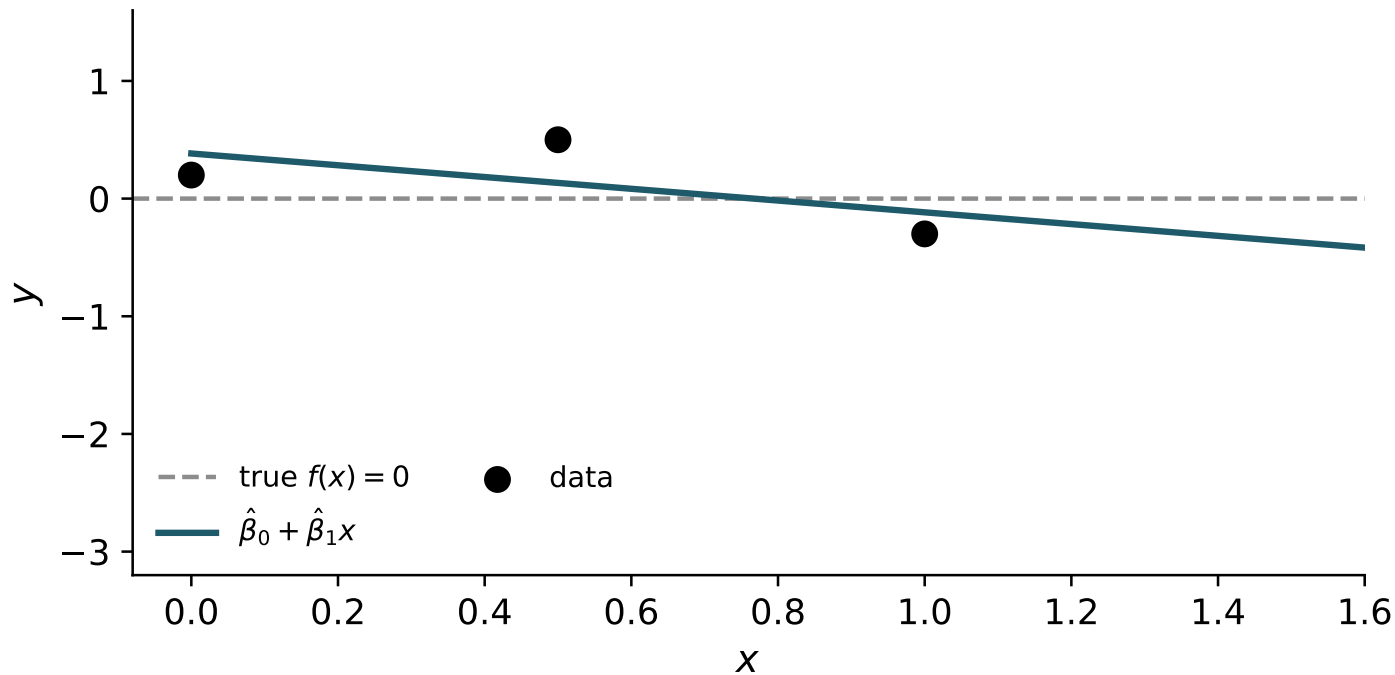


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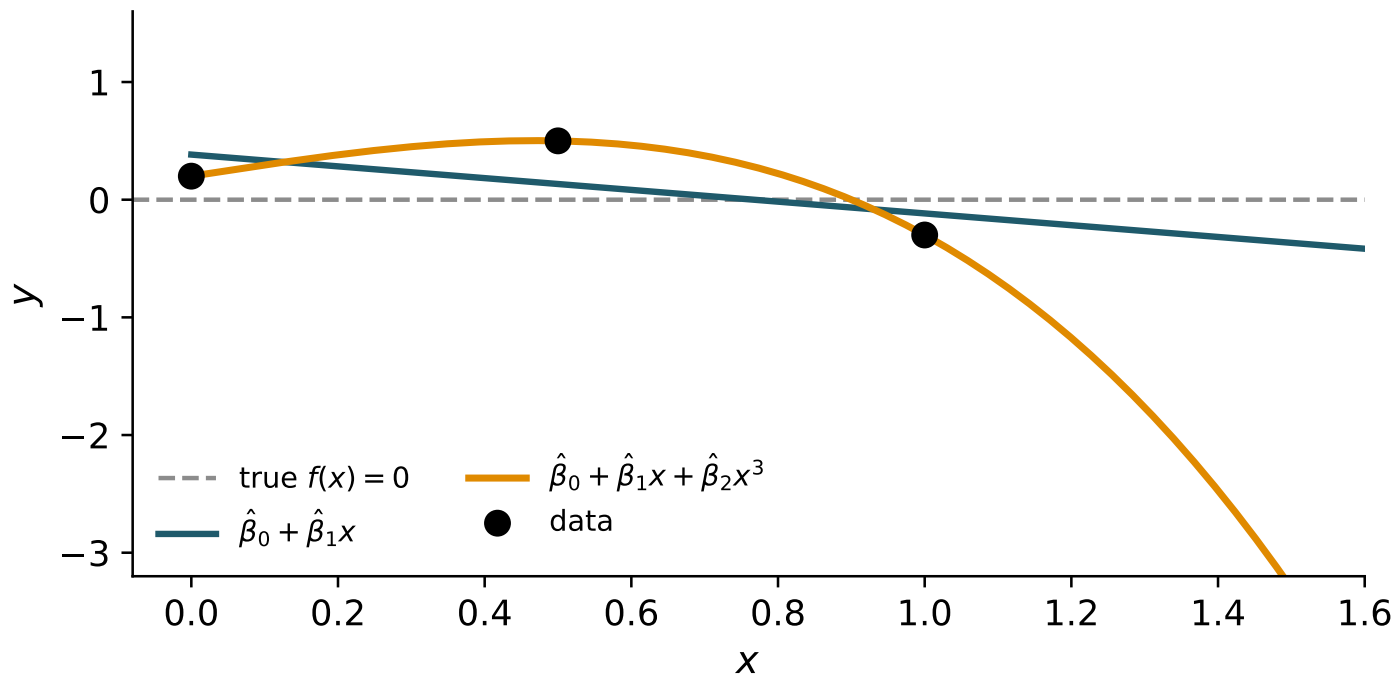
With only $\beta_0 + \beta_1 X$ (2 coefficients), the least-squares line cannot pass through all three points.

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